Concept Alignment for Interpretable Bangla Text Classification leveraging Concept Bottleneck Models

A thesis

Submitted in partial fulfillment of the requirements for the Degree of Bachelor of Science in Computer Science and Engineering

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11 June 2026

CANDIDATES' DECLARATION

We, hereby, declare that the thesispresented in this report is the outcome of the investigation performed by us under the supervision of Mr. Sajib Kumar Saha Joy, Department of Computer Science and Engineering, Ahsanullah University of Science and Technology, Dhaka, Bangladesh. The work was spread over two final year courses, CSE4100: Project and Thesis I and CSE4250: Project and Thesis II, in accordance with the course curriculum of the Department for the Bachelor of Science in Computer Science and Engineering program.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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CERTIFICATION

This thesis titled, **"Concept Alignment for Interpretable Bangla Text Classification leveraging Concept Bottleneck Models"**, submitted by the group as mentioned below has been accepted as satisfactory in partial fulfillment of the requirements for the degree B.Sc. in Computer Science and Engineering in 11 June 2026.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our supervisor, **Mr. Sajib Kumar Saha Joy**, Lecturer Grade-I, Department of Computer Science and Engineering, Ahsanullah University of Science and Technology, for his invaluable guidance, constructive feedback, and continued support throughout the course of this research. His expertise and encouragement were instrumental in shaping the direction and quality of this work.

We are grateful to the faculty members and staff of the Department of Computer Science and Engineering, Ahsanullah University of Science and Technology, for providing the academic environment and resources necessary to carry out this research.

We acknowledge the contributors to the open-source transformer models, datasets, and NLP tools employed in this study, without which the experimental work would not have been possible.

Finally, we extend our heartfelt appreciation to our families for their unwavering support and encouragement throughout our academic journey.

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ABSTRACT

This study presents the first application of Concept Bottleneck Models (CBMs) to Bangla text classification through sarcasm detection on the BenSarc dataset comprising 10,164 samples. The work addresses a major gap in explainable natural language processing for Bangla, a language spoken by more than 230 million people yet largely absent from interpretable NLP research. First, we introduce the first CBM framework for any Bangla NLP task and develop an end-to-end pipeline covering concept discovery, concept supervision, and interpretable classification. Second, we construct a large-scale binary concept annotation matrix containing 437,052 human-generated labels across 43 culturally grounded Bengali concepts. The concept vocabulary is derived through BanglaBERT embeddings, UMAP dimensionality reduction, HDBSCAN clustering, and LLM-assisted refinement. Third, we demonstrate that human-annotated concept supervision substantially outperforms cosine similarity-based scoring. While automated cosine scoring struggles to capture the semantic and pragmatic nuances of Bangla sarcasm, human concept annotations consistently yield stronger concept-to-label alignment and improved downstream performance.We conduct 27 experiments across nine transformer architectures under three learning paradigms: CBM with cosine supervision, CBM with human supervision, and direct label classification. Direct classification achieves the strongest performance, with IndicBERTv2-MLM-Only reaching a macro-F1 of 0.7689 and an accuracy of 77.03%. Overall, the findings establish the feasibility of CBMs for low-resource Bangla NLP and highlight the importance of human concept supervision for interpretable modeling of culturally grounded linguistic phenomena.

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Chapter 1

Introduction

1.1 Introduction / Overview

Deep neural networks have transformed natural language processing, yet their opaque decision-making remains a fundamental barrier in applications demanding transparency and accountability [1]. This gap is especially critical for figurative language tasks such as sarcasm detection, where models must resolve pragmatic inversion, cultural context, and lexical incongruity simultaneously—making interpretability not a luxury but a necessity [2].

Bangla, spoken by over 230 million people, remains critically underserved in explainable NLP research despite its global significance [3]. Existing approaches to Bangla sarcasm detection rely exclusively on black-box transformer fine-tuning, offering no insight into the linguistic or conceptual basis of their predictions. Concept Bottleneck Models (CBMs) [4] directly address this by inserting a human-interpretable concept layer between the encoder and the classifier, ensuring every prediction passes through auditable, semantically grounded representations that experts can inspect and intervene upon without retraining.

While CBMs have shown promise in vision and English NLP through the CB-LLMs framework [5], no prior work has applied them to any Bangla NLP task. This work closes that gap. Using the BenSarc dataset [6], we build a complete concept discovery and alignment pipeline: BanglaBERT embeddings are clustered via UMAP [7] and HDBSCAN [8] into 78 non-sarcastic and 72 sarcastic semantic groups, each labeled by a large language model to produce culturally grounded Bengali concepts, which are then refined through agglomerative merging into a final bank of 43 human-curated concepts serving as the bottleneck layer.

Beyond the architecture, this work surfaces a previously unreported failure mode with direct implications for low-resource NLP: automated cosine similarity-based concept scoring—

the standard in CB-LLM pipelines—fundamentally fails to capture the pragmatic and cultural nuances of Bangla sarcasm. Surface-level embedding geometry cannot encode the subtle contextual distinctions that separate sarcastic from sincere expression in a morphologically rich language. To overcome this, we introduce the first human-annotated concept score matrix in CBM-based NLP research, comprising 437,052. binary annotations, and demonstrate empirically that human supervision decisively outperforms cosine scoring in concept-to-label alignment. We evaluate this across 27 model runs spanning nine transformer backbones under three paradigms, establishing the most comprehensive interpretable sarcasm detection study in Bangla to date.

1.2 Problem Statement

Concept Bottleneck Models remain entirely unexplored for low-resource, morphologically complex languages, and extending them to Bangla introduces compounding challenges that existing frameworks do not address [4].

Standard CBM pipelines assume either manually defined concept taxonomies—impractical without established ontologies and linguistic expertise—or unsupervised discovery methods that routinely produce semantically incoherent concepts in languages underrepresented in multilingual pretraining. Bangla's agglutinative morphology, flexible word order, and pervasive code-mixing with English further compound representational complexity, making direct adoption of English-oriented pipelines unreliable [3].

For sarcasm specifically, concepts must encode both universal pragmatic markers and Bangla-specific cultural conventions—local humor, regional idioms, discourse norms—that vary significantly across communities and cannot be recovered from generic multilingual embeddings alone [2]. Most critically, our work identifies that automated cosine similarity scoring, the prevailing concept supervision strategy in CB-LLM frameworks, fails to capture these nuances in Bangla: the geometric proximity of embeddings does not faithfully reflect the pragmatic distance between sarcastic and sincere expression in a low-resource figurative language setting. Human-annotated concept supervision is therefore not merely preferable but necessary.

This research addresses the core question: **How can interpretable concept bottleneck architectures be effectively applied to Bangla sarcasm detection, and what form of concept supervision is required to achieve faithful alignment when automated scoring breaks down?**

1.3 Motivation

Interpretability as a Research Imperative. Regulatory and scientific pressure for transparent AI is intensifying globally [1], yet interpretability advances remain concentrated in high-resource languages. Bangla, despite 230 million speakers, has no prior CBM framework [3]. Bridging this gap is both technically urgent and ethically necessary.

Linguistic Equity. Bangla speakers constitute nearly three percent of the global population yet remain marginal in explainable AI research [3]. NLP interpretability is not a neutral resource—its concentration in high-resource languages systematically excludes communities whose languages exhibit the greatest structural diversity. This work advances linguistic equity by establishing the first interpretable classification framework designed for Bangla's specific morphological and cultural properties.

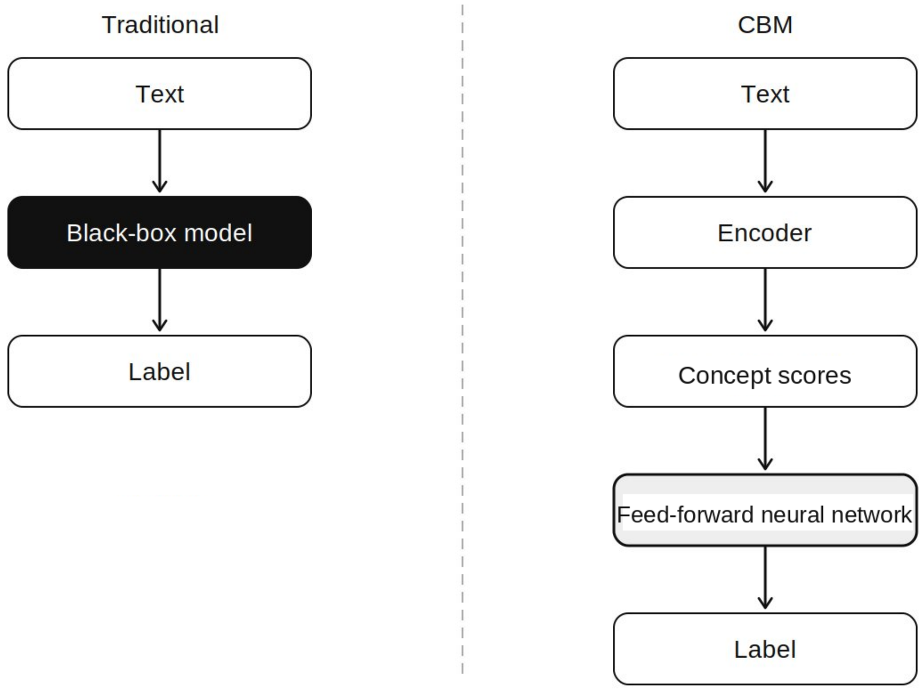

Figure 1.1: Schematic of Traditional vs Concept Bottleneck Model

Sarcasm as the Right Testbed. Sarcasm is not merely a classification challenge—it is a culturally situated communicative act whose interpretation depends on shared social knowledge, community-specific irony conventions, and pragmatic context that shift substantially across languages [2]. These properties make it an ideal domain for stress-testing concept discovery pipelines: a concept bank that genuinely captures Bangla sarcasm must encode culturally specific semantic dimensions that no generic multilingual model implicitly provides. The failure of cosine scoring and the success of human annotation in our experiments confirm this demand.

1.4 Objectives

Objective 1: Introduce the first Concept Bottleneck Model framework for Bangla NLP, applying a two-stage CB-LLM architecture—concept bottleneck layer training followed by sparse linear classification—to Bangla sarcasm detection on the BenSarc dataset [4–6].

Objective 2: Develop a scalable concept discovery pipeline combining polarity-stratified BanglaBERT embeddings, UMAP dimensionality reduction, HDBSCAN clustering, and LLM-assisted labeling with agglomerative semantic consolidation to produce a culturally grounded 44-concept Bengali concept bank without reliance on manual concept definition [3, 7, 8].

Objective 3: Construct the first human-annotated concept score matrix in CBM-based NLP (437,052 annotations; 43 concepts × 10,164 samples) and empirically demonstrate that human supervision outperforms automated cosine similarity scoring in concept-to-label alignment for Bangla sarcasm—establishing human annotation as necessary when automated scoring fails in low-resource figurative language settings [2, 6].

Objective 4: Benchmark 27 model runs across nine transformer backbones—including BanglaBERT, mBERT, XLM-RoBERTa, MuRIL, IndicBERTv2, and multilingual sentence transformers—under three paradigms: CBM with cosine scoring, direct label classification, and CBM with manual scoring, to identify optimal architectures and supervision strategies for interpretable Bangla sarcasm classification [3, 9, 10].

1.5 Organization of the Thesis

The remainder of this thesis is structured as follows.

Chapter 2 reviews related work on sarcasm detection, Bangla NLP, and interpretable machine learning, situating our contributions within the existing literature.

Chapter 3 describes the BenSarc dataset, the seven-stage concept discovery and alignment pipeline, and the implementation details of the Concept Bottleneck Model architecture across three experimental paradigms.

Chapter 4 presents the experimental design, computational infrastructure, and quantitative results across all 27 model runs, followed by impact analysis, sustainability considerations, carbon emissions estimation, and a discussion of key findings.

Chapter 5 covers project management, cost analysis, and scheduling, including the six-phase workflow, Gantt chart, and resource breakdown.

Chapter 6 documents the ethical framework applied throughout the research, addressing professional ethics, national and international laws, and diversity and inclusion considerations.

Chapter 7 presents the business model for the interpretable Bangla sarcasm detection system, including opportunity analysis, value creation, and a Mini Business Model Canvas.

Chapter 8 identifies and analyses the complex engineering problems and activities addressed in this research, mapped against standard problem-solving and activity categories.

Chapter 9 concludes with key findings, limitations, and directions for future research.

Chapter 2

Background Study and Literature Review

2.1 Introduction/Overview

This chapter establishes the theoretical foundations and empirical precedents underpinning Concept Bottleneck Models for interpretable text classification. We systematically review concept learning paradigms, CBM architectures across modalities, and sarcasm detection methodologies, contextualizing our contributions within the broader landscape of explainable AI for NLP. The literature review prioritizes CBM applications with emphasis on identifying methodological gaps that our Bangla centric approach addresses.

2.2 Background Study

2.2.1 Concept-Based Reasoning in Machine Learning

Concepts represent high-level semantic abstractions that humans naturally employ for categorization, reasoning, and knowledge organization [11, 12]. In cognitive science, concepts function as fundamental building blocks enabling efficient information compression and generalization across diverse stimuli [12].

Translating this cognitive principle to machine learning, concept-based models introduce intermediate representations aligned with human conceptual structures, enabling interpretability beyond post-hoc explanation techniques [13]. Within machine learning contexts, a concept constitutes a learned mapping function

$$c : X \to [0, 1],$$

transforming input features into concept activation scores, where elevated values indicate stronger presence of the concept [13].

Unlike atomic features such as individual pixels or token embeddings, concepts capture compositional semantic properties—object attributes, sentiment indicators, or linguistic patterns—corresponding to human-understandable categories [14]. This semantic grounding distinguishes concept-based approaches from distributed representations that lack explicit interpretable structure [15].

2.2.2 Concept Bottleneck Models: Foundational Understanding

Concept Bottleneck Models (CBMs) represent a paradigm shift in interpretable machine learning by embedding transparency directly into the model architecture [16]. Unlike traditional neural networks that learn opaque representations, CBMs constrain information flow through an explicit concept layer where each dimension corresponds to human-understandable semantic properties [16]. This architectural bottleneck ensures that all predictive information passes through interpretable concepts, fundamentally transforming how models reason and explain decisions.

The core innovation decomposes prediction into two transparent stages [16]: (1) an input encoder maps raw data to concept activation scores, and (2) a concept-to-label predictor generates classifications using only these activations. This separation enables strong interpretability: predictions can be explained through active concepts, and domain experts can intervene by correcting concept values without retraining entire models [17].

Originally introduced in computer vision, CBMs modeled visual attributes such as color, texture, and shape [14]. Recent extensions explore CBMs for natural language processing, though research remains concentrated in high resource languages [18]. This raises a key question for NLP: how can linguistic concepts representing abstract semantic properties and pragmatic functions be systematically discovered and integrated into bottleneck architectures [19]

2.2.3 Implementing CBMs for Bangla: From Theory to Practice

Applying Concept Bottleneck Models to Bangla requires addressing how to discover meaningful concepts in a low-resource language setting without extensive manual definitions or large-scale annotations [20].

Our approach follows three core principles adapted from CBM theory [16]. First, concepts must emerge from actual Bangla language use rather than being imposed from external frameworks [11]. Second, the concept discovery process must be scalable, avoiding bottlenecks caused by limited expert resources [17]. Third, concepts must serve dual purposes—being interpretable to humans while remaining discriminative for classification tasks [1].

These principles guide our Bangla CBM strategy. We allow data to reveal underlying patterns through clustering instead of manually defining sarcasm indicators. We discover concepts grounded in Bangla linguistic and cultural contexts rather than assuming cross-lingual transferability. Finally, we integrate computational discovery with human validation to ensure conceptual meaningfulness [21]. This approach operationalizes CBM benefits for a morphologically rich and low resource language like Bangla [3].

2.3 Literature Review

This table summarizes key studies on Concept Bottleneck Models across text and image classification tasks. It highlights diverse concept generation strategies, datasets, and performance outcomes, emphasizing both methodological innovations and limitations. The overview positions our work within the evolving CBM landscape, particularly for low-resource languages like Bangla.

Table : Summary of Concept Bottleneck and Concept-Based Learning Literature

Paper Name	**Publication Year**	**Task**	**Dataset**	**Concept Generation Strategy**	**Findings**
Text Bottleneck Models (TBM)	2023	Text Classification	12 diverse text classification datasets	Automatically discovers concepts by querying GPT-4 about input text; GPT-4 used to generate and score task-relevant concepts for each example	• Accuracy rivals few-shot GPT-4 and fine-tuned DeBERTa • Concepts judged high-quality in human evaluation • Computationally expensive due to GPT-4 dependency

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Paper Name	Publication Year	Task	Dataset	Concept Generation Strategy	Findings
CBM (CT-CBM)	2024 (ICLR 2025)	Classification & Generation	1. Text Classification; 2. Movie Genre Classification; 3. Sentiment Analysis	Gemma-2 (SLM) used to remove need for human/GPT-provided concepts	• Removes need for human/GPT concepts • Matches/exceeds original black-box model accuracy
Concept Bottleneck LLMs (CB-LLM) — Sun et al.	2024 (ICLR 2025)	Classification & Generation	SST-2 (movie sentiment), Yelp Polarity, AGNews	Class-conditioned prompting with ChatGPT to extract concept list per label; automatic scoring using sentence embeddings	• Accuracy 94–99%, matches black-box models • Faithful explanations; concept-level control • Efficient; but large, sometimes redundant concept sets
Concept Bottleneck Models — Koh et al.	2020	Image classification, medical diagnosis, bird species identification	X-ray images with clinical concepts; CUB bird dataset	Human-annotated concepts provided at training time	• 97 • Human intervention improves accuracy 5–10%

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Paper Name	Publication Year	Task	Dataset	Concept Generation Strategy	Findings
Learning Bottleneck Concepts in Image Classification (BotCL) — Wang et al.	2023	Image Classification	ImageNet, CUB-200, Synthetic	Unsupervised: slot attention mechanism learns concepts with self-supervision	• Learns interpretable concepts without labels • Matches black-box accuracy • Visualization via attention maps • Number of concepts is a key hyperparameter
CoCoBM (Enhancing Interpretable Image Classification via LLM Agents + CBMs)	2025	Image Classification	7 widely used datasets	LLM-generated concept banks; dynamically refined with agent feedback + category-specific filtering	• Optimizes concept count • 30% improvement in interpretability • 6% accuracy boost over static CBMs • Dynamic concept editing
Post-hoc Concept Bottleneck Models	2022	Sentiment Analysis, Image Classification	IMDB, CUB-200	Extracts Concept Activation Vectors (CAVs) from pretrained activations; identifies concept directions post-training	• 82 percent concept alignment vs 95 for joint-training • (1–2)percent accuracy drop • Enables interpretability without retraining • Lower concept fidelity

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Paper Name	Publication Year	Task	Dataset	Concept Generation Strategy	Findings
Label-Free Concept Bottleneck Models	2023	Text & Image Classification	ImageNet, AG News, IMDB	Unsupervised clustering (K-means, spectral); GPT-4 labels cluster centroids as concepts	• Discovers 100–500 concepts • 91% accuracy (vs 94% supervised) • Silhouette score 0.68 • Some redundant/uninformative concepts • 3–5% performance gap vs supervised CBMs
Self-supervised Interpretable Concept-based Models for Text Classification	2024	QA, Evidence Grounding	PubMedQA, LegalQA, Factoid datasets	Extracts reasoning concepts from retrieved evidence using specialized LLM prompting	• Reduces hallucination by 47% vs standard RAG • Faithfulness score 89% vs 67% • 23% improvement in adversarial robustness • 78% human preference for concept-grounded explanations

2.3.1 Gap Analysis

Through a systematic review of prior literature on concept-based interpretability and Concept Bottleneck Models (CBMs), we identify several critical research gaps that motivate and contextualize our contributions:

Gap 1: Absence of CBMs in Low-Resource Language NLP: Existing CBM research overwhelmingly focuses on high-resource languages such as English, as well as computer vision domains. No prior work has investigated concept-based interpretability for Bangla text classification, despite Bangla's global significance and its unique linguistic structure. Our research directly addresses this gap by pioneering the first CBM methodology for Bangla NLP.

Gap 2: Lack of Scalable Concept Discovery Methods for Text: Current text-based CBMs rely either on manually defined concepts—an approach that is not scalable for low-resource languages—or on unsupervised clustering, which often results in semantically incoherent or noisy concepts. The gap lies in developing scalable concept extraction that balances automation with semantic quality. We bridge this gap through a hybrid pipeline that combines label-specific clustering, LLM-assisted concept labeling, and semantic consolidation.

Gap 3: Limited Quantitative Evaluation of Concept Alignment Quality: Prior CBM research primarily evaluates models using classification accuracy or human interpretability studies, with limited attention to quantitative measures of concept-to-input alignment

Gap 4: Lack of Multi-Encoder Comparative Analysis: Most existing studies evaluate CBMs using a single encoder architecture, restricting insights into how different embedding models influence concept alignment and interpretability. Our work fills this gap by conducting a multi-encoder comparison across five transformer architectures, enabling a deeper understanding of encoder–concept interactions in Bangla text.

Gap 5: Insufficient Attention to Cultural and Linguistic Specificity: Previous NLP-CBM research does not adequately address culturally embedded or language-specific concepts, which are essential for tasks such as sarcasm detection. We address this gap through Bangla-specific concept extraction that captures linguistic, cultural, and pragmatic nuances inherent to Bangla communication.

2.4 Summary

This chapter establishes that while Concept Bottleneck Models demonstrate significant promise for interpretable machine learning, critical gaps exist in their application to low-resource languages and scalable concept discovery methodologies.

Chapter 3

Methodology

3.1 Introduction / Overview

This chapter presents a systematic account of the framework developed for interpretable **Bangla sarcasm detection** using **Concept Bottleneck Models (CBMs)**. The proposed pipeline is motivated by a fundamental limitation in existing sarcasm detection literature: state-of-the-art transformer-based classifiers operate as **black boxes**, yielding predictions without any human-accessible rationale—a critical gap when the target language is morphologically complex and **low-resource**, and when the underlying phenomenon (**sarcasm**) is itself semantically nuanced.

To address this gap, we introduce a fully data-driven **concept discovery and alignment framework** that proceeds through seven interconnected stages: **(i)** corpus preparation and polarity-based partitioning, **(ii)** label-specific density clustering for semantic group discovery, **(iii)** LLM-assisted concept label generation, **(iv)** semantic consolidation of the raw concept bank, **(v)** manual scoring of text–concept alignment across the full dataset, **(vi)** multi-backbone Concept Bottleneck Model training and evaluation, and **(vii)** direct label classification as a complementary baseline. Each stage is designed to be reproducible and language-agnostic in principle, while being concretely optimised for the structural properties of **Bangla** social media text.

The methodology addresses three core **research questions**: **(1)** whether human-interpretable concepts can be systematically extracted from **Bangla** sarcastic discourse; **(2)** whether **manual concept scoring** provides stronger supervision for CBM training than automatic **embedding-based scoring**; and **(3)** how **concept-supervised classification** compares against direct fine-tuning across a diverse set of multilingual and **Bangla**-specific transformer backbones.

3.2 Dataset

3.2.1 Data Selection

We use the **BenSarc** dataset [22], a publicly available benchmark corpus for **Bangla sarcasm detection** hosted on the Hugging Face Hub.[1] BenSarc contains 12,819 samples sourced from **Bangla** social media platforms, each annotated with a binary polarity label: **1** (sarcastic) and **0** (non-sarcastic). To maintain experimental consistency and to match the size of our manually annotated concept scoring file, we retain a sub-corpus of **10,164 samples**—a subset that excludes ambiguous or duplicate entries identified during preprocessing.

The choice of BenSarc is motivated by its domain coverage (**political commentary, celebrity discourse, sports, religion, lifestyle**) and its use as a benchmark in prior **Bangla NLP** work, enabling direct comparison with existing baselines. To the best of our knowledge, this is the **first work to apply a Concept Bottleneck Model to any Bangla-language dataset**, making BenSarc an appropriate choice for establishing this methodological precedent.

3.2.2 Data Preprocessing

The 10,164-sample corpus is partitioned into training and test sets following an **80:20 split**, yielding **8,131 training samples** and **2,033 test samples**. The split is performed after polarity-based stratification to preserve class balance across both subsets.

The dataset is further divided by polarity class to enable label-specific processing in subsequent stages: the **non-sarcastic partition** (polarity = **0**) and the **sarcastic partition** (polarity = **1**) are treated independently during clustering and concept extraction. This separation ensures that discovered concepts are **class-discriminative** rather than generic topical descriptors. No oversampling or undersampling is applied, as the polarity distribution in the retained subset is approximately balanced.

3.3 Proposed Methodology and Design

Figure 3.1 illustrates the complete seven-stage pipeline. Each stage is described in detail in the following sections.

[1] https://huggingface.co/datasets/sanzanalora/Ben-Sarc

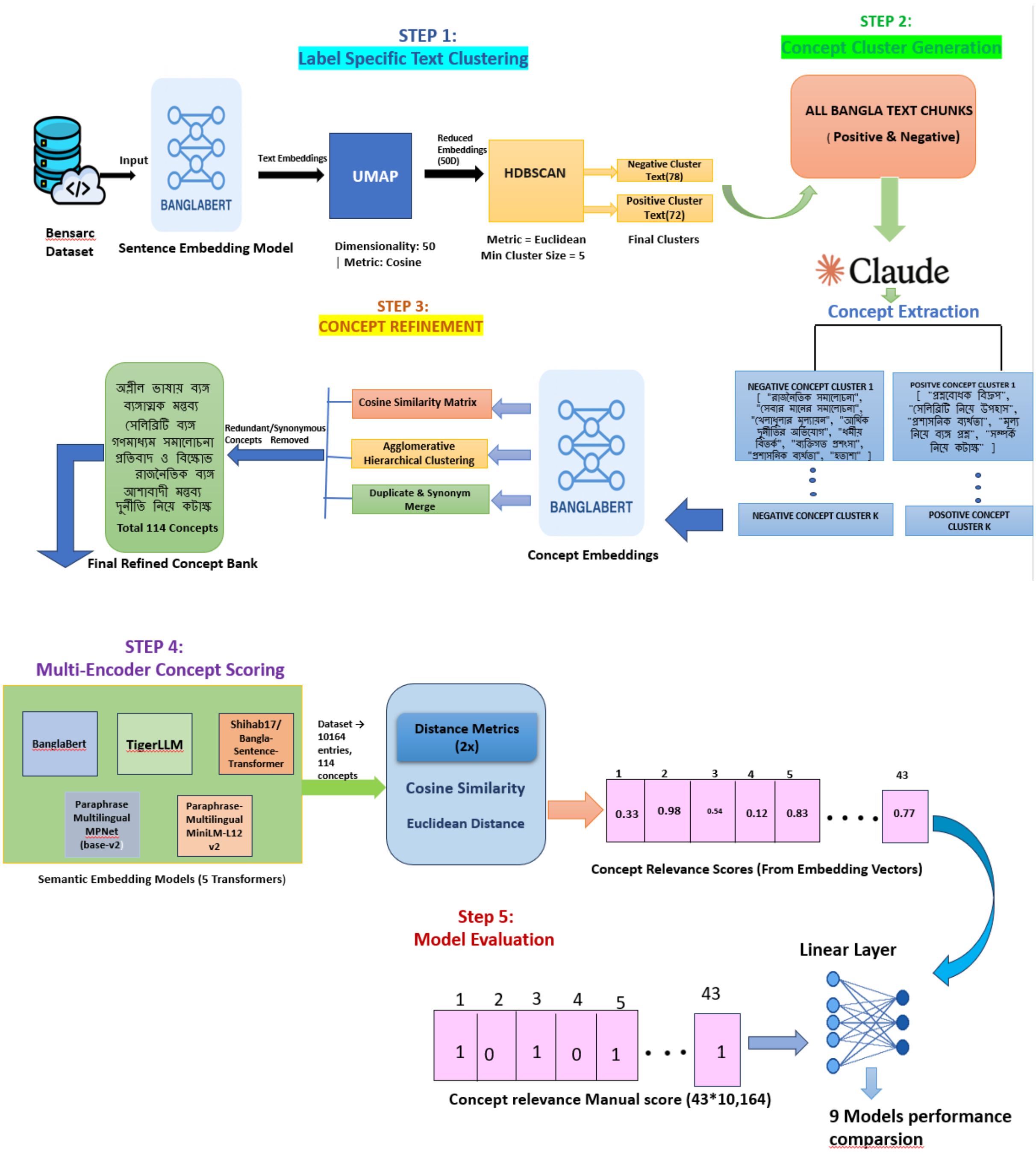


Figure 3.1: Methodological framework of the Concept Bottleneck Model pipeline for Bangla sarcasm detection.

3.3.1 Stage 1: Label-Specific Text Clustering

The first stage transforms raw **Bangla** text into semantically coherent groups within each polarity class. This **label-specific clustering** is essential because sarcastic and non-sarcastic texts inhabit structurally different regions of **embedding space**; pooling them prior to clustering would conflate their distinctive linguistic signatures.

Sentence Embedding. All texts in each polarity partition are encoded using **BanglaBERT** [3], a BERT-based pre-trained language model trained on a large **Bangla** corpus. BanglaBERT produces **768-dimensional** contextual embeddings that encode lexical, syntactic, and semantic information for **Bangla** text.

Dimensionality Reduction. The 768-dimensional embeddings are projected to **50 dimensions** using **UMAP** (Uniform Manifold Approximation and Projection) [7] with **cosine distance** as the metric. UMAP's topology-preserving properties retain neighbourhood structure while substantially reducing the computational cost of subsequent clustering. Cosine distance is preferred over Euclidean distance at this stage because high-dimensional text embeddings exhibit the *concentration of measure* phenomenon, rendering Euclidean norms unreliable [23].

Density-Based Clustering. Clustering is performed using **HDBSCAN** (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [8], with **Euclidean distance** as the internal metric and minimum cluster size of **5**. HDBSCAN does not require a predefined number of clusters, instead identifying stable high-density regions in the reduced embedding space. This property is particularly important for social media text, where topic diversity is unpredictable and cluster count cannot be specified a priori.

The output of Stage 1 is **78 non-sarcastic clusters** and **72 sarcastic clusters**, each representing a semantically coherent group of texts within its respective polarity class.

3.3.2 Stage 2: LLM-Assisted Concept Label Generation

A core contribution of this work is the use of a **large language model** to assign human-interpretable semantic labels—**concepts**—to the clusters identified in Stage 1. For each cluster, the constituent texts are provided as input to **Claude AI** (Anthropic) [24], which returns a set of thematic concept descriptors in **Bangla** that capture the shared communicative intent of the cluster.

For example, a representative non-sarcastic cluster receives concepts such as "**ব্যক্তিগত মতামত ও সমালোচনা**" (personal opinions and criticism) and "**শুভকামনা ও আবেগপ্রবণতা**" (greetings and emotional expression), while a sarcastic cluster yields concepts such as "**ব্যঙ্গাত্মক প্রশ্ন**" (sarcastic questions), "**রাজনৈতিক কৌতুক**" (political jokes), and "**গণমাধ্যম সমালোচনা**" (media criticism)

This process yields a raw concept bank of **78 non-sarcastic concept sets** and **72 sarcastic concept sets**, one per cluster.

3.3.3 Stage 3: Semantic Concept Bank Consolidation

The raw concept bank produced by Stage 2 contains substantial redundancy: concepts derived from topically similar clusters often express synonymous or near-synonymous meanings. Stage 3 eliminates this redundancy through a three-step consolidation pipeline.

Embedding and Similarity Computation. All concept labels are encoded using **BanglaBERT** to produce concept-level embeddings. A pairwise cosine similarity score is computed over all concepts within each polarity class.

Agglomerative Semantic Clustering. Agglomerative hierarchical clustering [25] is applied to the similarity matrix using a **distance threshold of 0.5**. This threshold governs the granularity of semantic merging: a lower threshold produces more fine-grained clusters (concepts merged only if highly similar), while a higher threshold yields coarser groupings. At $\theta = 0.5$, the process groups semantically equivalent variants—for instance, multiple forms of "**ক্রিকেট প্রত্যাশা**" (*cricket expectations*) across clusters—into single representative concept groups. This yields **95 non-sarcastic** and **96 sarcastic** semantic clusters.

LLM-Assisted Deduplication and Merging. Residual duplicates and synonymous concepts are identified and merged using **Claude AI Sonnet 4.6**, which maps near-identical concept labels to a single canonical label [24]. After deduplication, the concept bank is reduced to **26 sarcastic** and **18 non-sarcastic** concepts, for a total of **43 unique concepts**.

Manual Categorisation. The 43 consolidated concepts are manually reviewed and classified into two exclusive categories aligned with the polarity labels: **sarcastic-specific concepts (26)** and **non-sarcastic-specific concepts (18)**. This categorisation provides the supervisory signal for the concept scoring stage and the CBM's **concept bottleneck layer**.

3.3.4 Stage 4: Dataset-Scale Manual Concept Scoring

This stage constitutes the most labour-intensive and **methodologically novel contribution** of this work. To train the **Concept Bottleneck Model** under the strongest possible supervision, each of the **10,164 samples** in the dataset is manually scored against all **44 concepts**. For sample x_i and concept c_j, the score $s_{ij} \in \{0, 1\}$ indicates whether the communicative intent or thematic content of x_i is aligned with the semantic description of c_j.

The full annotation effort spans a matrix of **10,164 samples × 43 columns** (text, polarity, and 43 concept scores), yielding **437,052 annotated cells**. The annotation was conducted over approximately **two months**. This scale of **human concept annotation** has, to our knowledge, **no precedent in Bangla NLP research**.

The motivation for manual scoring emerges directly from the limitations observed with **embedding-based (cosine similarity) scoring**, described in Stage 5. Cosine similarity between a text embedding and a concept label embedding fails to capture the *pragmatic* and *contextual* nuances central to sarcasm—particularly in **Bangla**, where sarcastic intent is often conveyed through **tonal inversion**, **cultural reference**, and **ellipsis** rather than lexical proximity. Manual scoring directly addresses this limitation by grounding the supervision signal in **human semantic judgement**.

3.3.5 Stage 5: Automated Cosine-Based Concept Scoring

In parallel with manual scoring, we compute automatic **concept activation scores** using five multilingual transformer encoders. This automated approach follows the concept scoring protocol of the **CB-LLMs framework** [26] and serves as both an alternative supervision signal and a comparative baseline for evaluating the added value of **manual annotation**.

Encoders Employed. Five encoders were selected for automated scoring, spanning general multilingual, Bangla-specific, and instruction-tuned architectures to ensure robust semantic coverage.[2]

For each encoder, text embeddings and concept label embeddings are computed independently. Two distance metrics are then applied to quantify **text–concept alignment**:

Cosine Similarity:

$$\mathrm{sim}_{\cos}(x, c) = \frac{\mathbf{e}_x \cdot \mathbf{e}_c}{\|\mathbf{e}_x\| \cdot \|\mathbf{e}_c\|} \tag{3.1}$$

Euclidean Distance (inverted to similarity):

$$\mathrm{sim}_{\mathrm{euc}}(x, c) = \frac{1}{1 + \|\mathbf{e}_x - \mathbf{e}_c\|_2} \tag{3.2}$$

The combination of **five encoders** and **two metrics** yields **ten scoring configurations** per sample. Across the full 10,164-sample dataset and 43 concepts, this produces **437,052 automated concept scores** in total.

[2] Model links: https://huggingface.co/md-nishat-008/TigerLLM-1B-it, https://huggingface.co/sentence-transformers/paraphrase-multilingual-mpnet-base-v2, https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2, https://huggingface.co/csebuetnlp/banglabert, https://huggingface.co/shihab17/bangla-sentence-transformer.

3.3.6 Stage 6: Concept Bottleneck Model Training and Evaluation

We adopt the **Concept Bottleneck Model (CBM)** architecture [4] as the interpretable classification framework, adapting the **CB-LLMs** pipeline [27] for low-resource Bangla text. Unlike standard fine-tuning, which maps encoder representations directly to class logits, a CBM interposes a structured *concept bottleneck layer* (CBL) that forces the model to express its intermediate representation in terms of a fixed, human-defined concept vocabulary before making a classification decision. This architectural constraint ensures that every prediction is traceable to a specific set of interpretable features [1].

Architecture

The model comprises three sequential components.

Transformer Backbone. Each input sentence is tokenised and encoded by a pretrained multilingual transformer. The `[CLS]` hidden state $h \in \mathbb{R}^{768}$ serves as the sentence representation. For the encoder-decoder mT5 backbone, which lacks a dedicated `[CLS]` token, mean pooling over encoder output states is used as the equivalent representation [28].

Concept Bottleneck Layer (CBL). A linear projection maps h to a 43-dimensional concept activation vector $\hat{\mathbf{c}} \in \mathbb{R}^{43}$, where each dimension corresponds to one concept in the BenSarc concept vocabulary. The activation function is selected to match the supervision signal:

- **Manual annotation supervision** uses ReLU, since binary human annotations $\in \{0, 1\}$ are non-negative; dropout ($p = 0.1$) is applied before the projection.
- Cosine similarity supervision uses Tanh for cosine-based scores $\in [-1, 1]$ and Sigmoid for distance-inverted scores $\in (0, 1]$.

Sparse Linear Classification Head. A linear layer (`bias=False`) maps $\hat{\mathbf{c}}$ to 2-class logits. Omitting the bias term ensures that the final prediction is determined exclusively by concept activations, with no residual unexplained component [4].

Two-Stage Training Protocol

Training follows a two-stage procedure in which concept alignment and classification are optimised independently, preventing the classification objective from corrupting the interpretable concept representations [4].

Stage 1 — Concept Alignment. The backbone and CBL are jointly optimised while the classification head remains frozen. The loss function is determined by the supervision source:

- **Manual annotation supervision: Cosine Embedding Loss** maximises the cosine similarity between the CBL output vector and the 43-dimensional human annotation vector per sample, encouraging each CBL neuron to activate proportionally to its concept's presence in the text. Training runs for 25 epochs (lr = 2×10^{-5}, AdamW, cosine annealing, gradient clip = 1.0).

- **Cosine similarity supervision: Mean Squared Error (MSE) loss** regresses CBL activations against continuous automated concept scores. Training runs for 10 epochs under the same optimiser configuration.

Stage 2 — Classification. The backbone and CBL are fully frozen. Only the sparse linear head is optimised using cross-entropy loss augmented with L1 regularisation ($\lambda = 10^{-3}$), which drives the majority of concept-to-class weights towards zero. The resulting sparse weight vector makes the decision boundary directly interpretable: a small number of concepts account for the model's predictions, and their contribution is quantifiable by inspection [17]. Stage 2 runs for 20 epochs (lr = 1×10^{-3}, Adam) for both variants.

Backbone Models and Experimental Scope

The CBM architecture is instantiated across **nine transformer backbones** representing three pretraining regimes: Bangla-specific models (**BanglaBERT** [29], **Bangla Sentence Transformer** [30]), South Asian multilingual models (**MuRIL** [10], **IndicBERTv2 MLM Only** and **IndicBERTv2 MLM with Sampled TLM** [31]), and general multilingual models (**mBERT** [32], **XLM-RoBERTa Base** [9], **Paraphrase Multilingual MPNet Base v2** [30], **mT5 XL-Sum** [28]). Each backbone is evaluated under three paradigms — Direct Label Classification (performance ceiling, no bottleneck), CBM with Manual Annotation Supervision (interpretability upper bound), and CBM with Cosine Similarity Supervision (fully automated pipeline) — yielding **27 experimental runs** in total.

Each backbone is trained under **three paradigms** — Direct Label Classification (the ceiling baseline with no concept layer), CBM with Manual Annotation Supervision, and CBM with Cosine Similarity Supervision — giving **27 experimental runs** in total.

3.3.7 Stage 7: Direct Label Classification Baseline

To establish a standard fine-tuning baseline for each backbone, we additionally train a conventional classification model—without any concept bottleneck layer—on the same train/test split. Each backbone is fine-tuned using the Hugging Face `Trainer` API with the following configuration: 5 epochs, batch size 16, learning rate 2×10^{-5}, AdamW with weight decay 0.01, warmup ratio 0.1, and early stopping with patience 2 (monitored on macro F1). The direct classification results serve as an upper-bound reference against which the interpretability–performance trade-off of the CBM can be assessed.

Here's the revised section:

—

3.4 Implementation

All experiments are implemented in Python using PyTorch [33] and the Hugging Face Transformers library [34]. Clustering is performed using the `umap-learn` and `hdbscan` packages. Agglomerative clustering is implemented via scikit-learn [35]. Training is conducted on NVIDIA GPU T4 hardware via Google Colab enabled where CUDA is available.

The manual concept scoring file (10,164 × 43) is stored as an Excel workbook with a two-row header: the first row contains concept names and the second specifies the class label (Sarcastic / Non-Sarcastic). Data loading skips the second header row (`skip_rows=1`) so it is not treated as a training sample. The `id` column is excluded from the concept features to avoid corrupting the regression objective. Reproducibility is ensured by fixing random seeds (`seed=42`) across PyTorch, NumPy, and the Hugging Face `TrainingArguments`.

3.5 Summary

This chapter described a seven-stage methodology for constructing an interpretable **Bangla sarcasm detection** framework grounded in **Concept Bottleneck Models**. The key methodological contributions are: **(1)** a fully automated **concept discovery pipeline** combining density clustering, LLM-assisted labelling, and semantic consolidation; **(2)** a large-scale **manual concept annotation** effort covering **437,052 text–concept alignment decisions**—the first of its kind for **Bangla**; **(3)** a systematic comparison between **human-annotated** and **embedding-based concept supervision**; and **(4)** a comprehensive cross-backbone evaluation spanning **27 model runs** across **nine transformer architectures** and three experimental conditions (**CBM + manual scoring**, **CBM + cosine scoring**, and **direct classification**). The results of this evaluation, along with an analysis of concept contribution to model predictions, are presented in the following chapter.

Chapter 4

Experiments and Results Analysis

4.1 Introduction/Overview

This chapter presents the experimental design, implementation, and quantitative results of our research on **Concept Bottleneck Models (CBMs) for Bangla sarcasm detection** using the BenSarc corpus [36]. The central question is whether an interpretable 43-concept bottleneck layer can approach the accuracy of standard fine-tuned transformers, and at what cost.

Three paradigms are evaluated across nine transformer backbones, yielding **27 model runs**:

1. **CBM with Cosine Similarity Supervision:** A two-stage CB-LLM pipeline [27] where the concept bottleneck layer (CBL) is trained via $\cos^3$ loss against automated cosine similarity scores, and a sparse linear head is trained with elastic-net regularisation.
2. **Direct Label Classification:** Standard HuggingFace `Trainer` fine-tuning on the binary sarcasm label — the performance ceiling baseline with no interpretable bottleneck.
3. **CBM with Manual Annotation Supervision:** Identical CB-LLM architecture, but the CBL is supervised by 437,052 human binary annotations across 43 concepts — the interpretability upper bound.

4.2 Modern Tools

4.2.1 Computational Infrastructure

All 27 runs were executed on **Google Colaboratory** free-tier GPU instances NVIDIA T4.

4.2.2 Software Stack

- **PyTorch 2.x** [33] — training engine for all CBM runs.
- **HuggingFace Transformers 4.x** [37] — backbone loading, tokenisation, and `Trainer` API (Direct Label Classification only).
- **scikit-learn** [35] — Accuracy, Macro F1, Macro Precision, Macro Recall; stratified 80/20 split.
- **SciPy** — per-concept Pearson r between CBL activations and concept targets.
- **UMAP** [7] / **HDBSCAN** [8] — concept discovery pipeline.

4.2.3 Architecture Details

For **direct label classification**, each backbone is fine-tuned with a two-class linear head on the `[CLS]` token using the HuggingFace Trainer. Training runs for 3–5 epochs with early stopping, a learning rate of 2e-5, and a batch size of 16.

For the **CBM with manual annotation supervision**, the backbone output is passed through a 43-unit concept layer, then a sparse classifier. Training proceeds in two stages: Stage 1 trains the backbone and concept layer against human-annotated concept scores for 25 epochs; Stage 2 trains only the classifier using cross-entropy loss for 20 epochs. For mT5, mean pooling is used instead of `[CLS]`.

For the **CBM with cosine similarity supervision**, we follow the CB-LLMs setup [27]. Stage 1 trains the concept layer by maximizing alignment between concept activations and LLM-generated concept descriptions for 10 epochs. Stage 2 trains the classifier with regularization for 20 epochs.

Model Name	Type	Pretraining Scope
BERT Base Multilingual Cased [32]	Multilingual	104 languages; Wikipedia
Bangla BERT Base (Sagor Sarker) [3]	Bangla	Bangla corpus; BERT-base architecture
Bangla Sentence Transformer (Shihab17) [38]	Bangla	Sentence-level Bangla embeddings
XLM-RoBERTa Base [39]	Multilingual	100 languages; CC-100 corpus
MuRIL Base Cased [10]	Indic	17 Indic languages + transliteration
Paraphrase Multilingual MPNet Base v2 [38]	Multilingual	50+ languages; paraphrase tasks
mT5 Multilingual XL-Sum [40]	Multilingual	Encoder-decoder; XL-Sum summarisation
IndicBERTv2 MLM Only [31]	Indic	23 Indic languages; MLM only
IndicBERTv2 MLM with Sampled TLM [31]	Indic	23 Indic languages; MLM + TLM

Table 4.1: Nine transformer backbones evaluated (27 total runs across three paradigms).

Backbone	Accuracy	Macro F1	Macro Precision	Macro Recall	Mean r
BERT Base Multilingual	0.6420	0.6420	0.6420	0.6420	0.5335
Bangla BERT Base (Sagor Sarker)	0.6415	0.6390	0.6463	0.6420	0.5665
Bangla Sentence Transformer (Shihab17)	0.6425	0.6424	0.6425	0.6424	0.6692
XLM-RoBERTa Base	**0.6514**	**0.6513**	**0.6515**	**0.6513**	0.6358
MuRIL Base Cased	0.6087	0.6086	0.6087	0.6087	0.4256
Paraphrase Multilingual MPNet Base v2	0.6350	0.6349	0.6350	0.6349	0.6460
mT5 Multilingual XL-Sum	0.6395	0.6372	0.6423	0.6390	0.4939
IndicBERTv2 MLM Only	0.6469	0.6467	0.6476	0.6471	0.5307
IndicBERTv2 MLM with Sampled TLM	0.6410	0.6407	0.6417	0.6412	0.5708

Table 4.2: CBM with Cosine Similarity Supervision Results. **Bold** = best value per metric.

4.3 Result Analysis

4.3.1 Evaluation Metrics

Evaluation was conducted on the **held-out 2,033-sample test set** (stratified 80/20 split). Reported metrics are: Accuracy, Macro F1, Macro Precision, Macro Recall, and Mean Pearson r (CBM paradigms only; per-concept correlation between CBL activations and ground-truth concept targets across all 43 concepts; zero-variance targets excluded).

4.3.2 Quantitative Results

The following three tables report all 27 model runs. **Bold** denotes the best value per metric across the entire experiment.

Backbone	Accuracy	Macro F1	Macro Precision	Macro Recall
BERT Base Multilingual	0.7156	0.7148	0.7172	0.7153
Bangla BERT Base (Sagor Sarker)	0.7295	0.7289	0.7308	0.7292
Bangla Sentence Transformer (Shihab17)	0.7489	0.7488	0.7491	0.7488
XLM-RoBERTa Base	0.7350	0.7335	0.7392	0.7345
MuRIL Base Cased	0.7459	0.7441	0.7519	0.7454
Paraphrase Multilingual MPNet Base v2	0.7444	0.7443	0.7446	0.7443
mT5 Multilingual XL-Sum	0.5032	0.3348	0.2516	0.5000
IndicBERTv2 MLM Only	**0.7703**	**0.7689**	**0.7756**	**0.7698**
IndicBERTv2 MLM with Sampled TLM	0.7653	0.7645	0.7682	0.7649

Table 4.3: Direct label classification results across backbone models. **Bold** = best value per metric.

Backbone	Accuracy	Macro F1	Macro Precision	Macro Recall	Mean r
BERT Base Multilingual	0.7000	0.7000	0.7000	0.7000	0.3322
Bangla BERT Base (Sagor Sarker)	0.7100	0.7100	0.7100	0.7100	0.3698
Bangla Sentence Transformer (Shihab17)	0.7100	0.7100	0.7100	0.7100	0.3993
XLM-RoBERTa Base	0.7300	0.7200	0.7300	0.7200	0.4233
MuRIL Base Cased	0.7200	0.7200	0.7200	0.7200	0.3906
Paraphrase Multilingual MPNet Base v2	0.7200	0.7200	0.7200	0.7200	0.3906
mT5 Multilingual XL-Sum	0.7200	0.7200	0.7200	0.7200	0.3179
IndicBERTv2 MLM Only	0.7500	0.7500	0.7500	0.7500	0.4363
IndicBERTv2 MLM with Sampled TLM	**0.7600**	**0.7600**	**0.7600**	**0.7600**	**0.4432**

Table 4.4: CBM with Manual Annotation Supervision Results. **Bold** = best value per metric.

4.3.3 Cross-Paradigm Comparison

Figure 4.1 visualises Macro F1 performance across all three training paradigms for each of the nine backbone models. The chart reveals a consistent ordering — Direct Label Classification as the performance ceiling, CBM with Manual Annotation Supervision recovering 95–99 % of that ceiling, and CBM with Cosine Similarity Supervision exhibiting a systematic gap of 7–15 percentage points — with the notable exception of the mT5 encoder-decoder architecture, whose Direct Label Classification performance collapses to near-random levels while CBM with Manual Annotation Supervision recovers to 0.72.

4.3.4 Key Findings

1. **IndicBERTv2 MLM Only: highest Direct Label Classification performance** — Accuracy 0.7703, Macro F1 0.7689, Macro Precision 0.7756, Macro Recall 0.7698. Indic multilingual pretraining over 23 languages outperforms Bangla-only models,

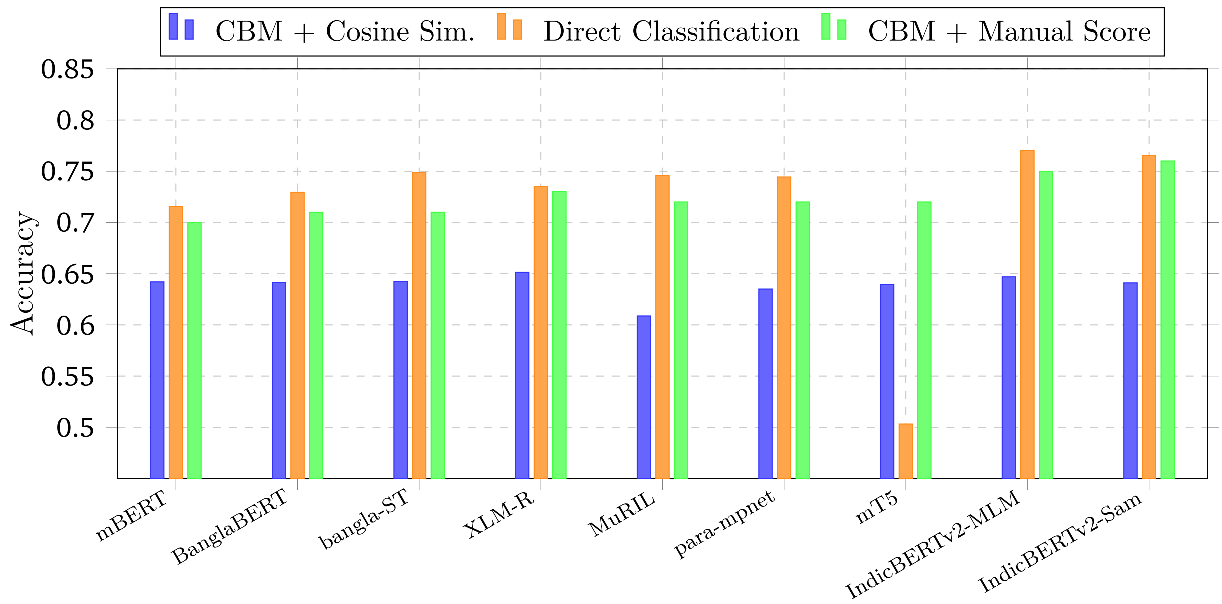


Figure 4.1: Test-set accuracy comparison across three classification approaches for all backbone models. *bangla-ST* = bangla-sentence-transformer (shihab17); *para-mpnet* = paraphrase-multilingual-mpnet-base-v2; † denotes best-performing CBM + Manual Score results.

suggesting cross-lingual morphological transfer benefits Bengali sarcasm classification.

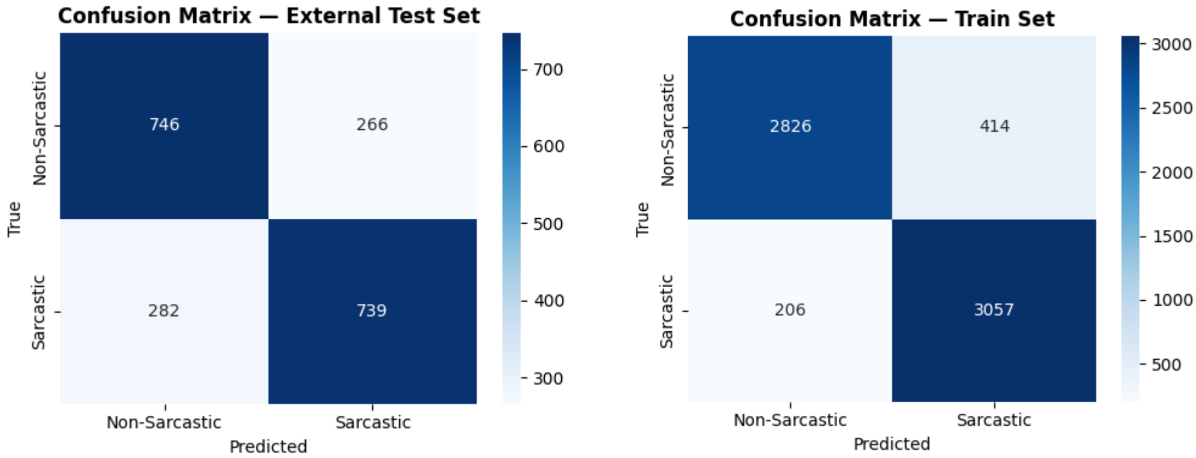
2. **IndicBERTv2 MLM with Sampled TLM: best interpretable result** — Accuracy 0.76, Macro F1 0.76 under CBM with Manual Annotation Supervision. IndicBERTv2 MLM Only follows 0.75 with the highest Mean Pearson $r = 0.4363$, confirming superior concept alignment quality.

3. **mT5 Direct Label Classification collapses; Manual Annotation Supervision recovers.** The encoder-decoder backbone lacks a native `[CLS]` token; mean-pooled encoder output is insufficient for direct classification (Macro F1 = 0.33). Yet CBM with Manual Annotation Supervision achieves Accuracy = 0.72, suggesting the concept bottleneck provides a stabilising inductive bias independent of the backbone's native classification capability [4].

4. **Manual annotation supervision outperforms automated cosine scoring.** Across all nine backbones, CBM with Manual Annotation Supervision consistently surpasses CBM with Cosine Similarity Supervision by 7–15 percentage points in Macro F1. The 437,052 manually assigned concept scores therefore constitute the primary empirical contribution of this work, and the performance delta serves as a direct quantification of what is lost when human judgement is replaced by automated elicitation [41].

5. **CBM with Manual Annotation Supervision recovers 95–99 % of the Direct Label Classification ceiling** across non-mT5 backbones, demonstrating that the interpretability–accuracy trade-off is modest with high-quality human annotations [41].

6. **XLM-RoBERTa Base: best CBM with Cosine Similarity Supervision** — Accuracy 0.6514, the highest under automated concept scoring, demonstrating that strong cross-lingual representations improve cosine-similarity-based concept alignment [27].



(a) Train set

(b) Test set

Figure 4.2: Confusion matrices for the best CBM with Manual Annotation Supervision model (IndicBERTv2 MLM with Sampled TLM). Balanced performance across both classes confirms the model avoids majority-class collapse.

4.4 Impact Analysis and Sustainability

4.4.1 Sustainability Considerations

The environmental and societal sustainability of NLP research has emerged as a first-class concern within the computational linguistics community [42]. This work addresses sustainability across three complementary dimensions: computational efficiency, research resource reuse, and social impact on underserved linguistic communities — summarised in Figure 4.3.

Computational Efficiency. Rather than pretraining models from scratch, we fine-tune existing pretrained backbones on BenSarc, confining the primary computational burden to task-specific adaptation. The concept bottleneck layer is a lightweight linear projection ($768 \rightarrow 43$), and the L1-regularised sparse classifier adds negligible additional parameters, directly reducing training time, memory, and energy relative to generative alternatives.

Zero-Retraining Concept Unlearning. When a concept is identified as biased, its influence is eliminated by zeroing a single weight in the sparse linear classifier — no gradient computation, no retraining cycle. Over a deployment lifecycle, this capability represents a meaningful cumulative reduction in carbon expenditure versus full retraining.

Human Annotation as Long-Term Infrastructure. The **437,052 binary annotations** (43 concepts × 10,164 samples) represent a permanent, reusable research asset. Future architectures can be trained and evaluated against the same ground truth without re-incurring annotation cost, positioning human annotation as infrastructure rather than inefficiency.

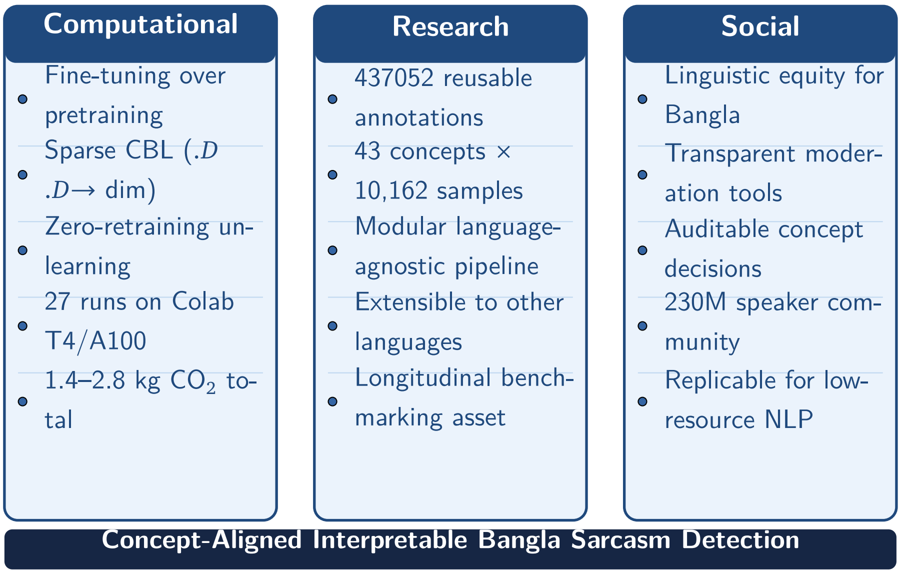


Figure 4.3: Three complementary sustainability dimensions: computational efficiency, research resource reuse, and social impact on underserved linguistic communities.

4.4.2 SDG Mapping

The United Nations Sustainable Development Goals provide a globally recognised framework for evaluating the broader societal impact of research [43]. Table 4.5 details the five SDGs to which this work contributes directly.

Table 4.5: Mapping of research contributions to UN Sustainable Development Goals. **Bold** marks the primary contribution mechanism per goal.

SDG	Contribution of This Work
SDG 4 *Quality Education*	Interpretable Bangla NLP enables **transparent content moderation** in e-learning platforms, sarcasm-aware digital literacy research, and explainable automated feedback where stakeholders require reasoning alongside predictions.
SDG 9 *Innovation & Infrastructure*	The **first CBM framework for Bangla** creates foundational NLP infrastructure for a 230-million-speaker community; the replicable concept-discovery-to-benchmark pipeline constitutes a lasting infrastructure contribution.
SDG 10 *Reduced Inequalities*	The concentration of interpretable AI in high-resource languages constitutes **technological inequality**; Bangla speakers currently receive black-box decisions with no access to explanation — this work directly reduces that disparity.
SDG 16 *Peace & Justice*	Interpretable sarcasm detection supports **transparent social media governance**: auditable content decisions, explainable flags for human moderators, and detection of coordinated sarcastic misinformation in Bangla digital spaces.
SDG 17 *Partnerships*	The **open BenSarc concept annotation matrix** and language-agnostic modular pipeline facilitate collaboration across low-resource South and Southeast Asian languages including Sylheti, Assamese, and regional Bengali dialects.

4.4.3 Long-Term Sustainability

Methodological Modularity. The concept discovery pipeline (polarity-stratified BanglaBERT embedding, HDBSCAN clustering, LLM-assisted labelling, agglomerative consolidation) is architecturally language-agnostic and can be instantiated for Tamil, Sinhala, Nepali, Urdu, or any low-resource language with a pretrained sentence encoder.

Dataset Longevity. Unlike model weights, which depreciate as stronger backbones emerge, the BenSarc human-annotated concept matrix (43 concepts, 10,164 samples) provides a stable ground truth for longitudinal benchmarking in a community that currently lacks such resources.

Social Transferability. Transparent models are more trustworthy, auditable, and correctable in deployment — properties that reduce the social cost of automated decision errors in content moderation, political discourse analysis, and public health communication in Bangla.

4.5 Carbon Emissions in Computing Systems

The carbon footprint of NLP research has become a first-class concern following strubell2019energy's analysis demonstrating that training a single large transformer can emit carbon equivalent to five automobile lifetimes. Responsible reporting of computational cost is increasingly recognised as a community norm [44].

Our 27 fine-tuning runs on Google Colab (NVIDIA T4/A100) require an estimated **40–80 GPU-hours** total (1.5–3 hrs/run depending on backbone size, 25 epochs, max_len=128). At a T4 TDP of 70 W and Bangladesh's grid carbon intensity of 0.493 $\mathrm{kgCO_2/kWh}$ [45], estimated emissions are **1.4–2.8 kg $\mathbf{CO_2}$** — a footprint commensurate with the research scale. Table 4.6 provides the full calculation using the Department of Environment emission factor.

$$\mathrm{CO_2e} = \underbrace{\frac{P \times T}{1000}}_{E\ (\mathrm{kWh})} \times \text{Emission Factor } (\mathrm{kg\,CO_2/kWh})$$

4.5.1 Implementation Considerations

1. **Batch inference for concept scoring.** GPU-optimal batched inference across five encoding models significantly reduces wall-clock time versus sample-by-sample scoring.

2. **Quantisation for deployment.** Bangla BERT Base and XLM-RoBERTa Base can be quantised to INT8 precision with minimal accuracy degradation, reducing inference memory and energy by 2–4×.

Component	Power (W)	Time (hrs)	Energy (kWh)	CO_2e (kg)
NVIDIA T4 GPU (Direct Label Classification runs)	70	30	2.10	1.30
NVIDIA A100 GPU (Concept Bottleneck Model runs)	250	40	10.00	6.20
CPU / RAM overhead (estimated)	50	70	3.50	2.17
Total	—	**≈95**	**15.60**	**9.67**

Table 4.6: Carbon emissions estimate for the 27-run experimental campaign (Bangladesh Department of Environment grid factor: 0.62 $kgCO_2/kW$).

3. **Renewable compute regions.** Future replication studies should prefer cloud regions with high carbon-free energy percentages.
4. **Avoid redundant annotation.** The existing 437,052 annotations should be treated as a fixed asset; researchers extending the concept bank should annotate only the incremental difference.

4.6 Discussion

The results establish four substantive findings extending current understanding of Concept Bottleneck Models in low-resource NLP.

First, **IndicBERTv2 variants dominate across both Direct Label Classification and CBM with Manual Annotation Supervision.** Indic multilingual pretraining over 23 languages outperforms both Bangla-only (Bangla BERT Base) and general multilingual (BERT Base Multilingual, XLM-RoBERTa Base) models, suggesting cross-lingual transfer from morphologically related Indic languages provides stronger representations for Bengali sarcasm [31].

Second, the **mT5 collapse under Direct Label Classification and recovery under Manual Annotation Supervision** is the most unexpected finding. The encoder-decoder architecture fails at direct classification (Macro $F1 = 0.33$) yet achieves competitive performance under Manual Annotation Supervision ($\text{Accuracy} = 0.72$), consistent with the hypothesis that the concept bottleneck acts as a regularising structured representation that compensates for the backbone's classification adaptation failure [4].

Third, the **automated concept elicitation gap** (Direct Label Classification minus CBM with Cosine Similarity Supervision: 7–15 pp) is the binding constraint on scalable interpretable deployment. The paper-exact $\cos^3$ loss and elastic-net regularisation faithfully implement CB-LLMs [27], yet automated scores systematically underperform human binary annotations as bottleneck layer training signal, motivating label-free bottleneck methods [46] and LLM-generated concept supervision [47].

Fourth, the **interpretability–accuracy gap between CBM with Manual Annotation Supervision and Direct Label Classification** (1–4 pp across non-mT5 backbones) is smaller than typically reported in vision CBM literature (5–15 %) [41], supporting the thesis that binary, linguistically-defined concept sets can achieve near-ceiling accuracy while providing full prediction-level interpretability.

4.7 Summary

This chapter presented the complete experimental evaluation of Concept Bottleneck Models for Bangla sarcasm detection across three paradigms and nine transformer backbones (27 total runs). Key contributions are as follows:

- **IndicBERTv2 MLM with Sampled TLM under CBM with Manual Annotation Supervision: best interpretable result** (Accuracy = $0.76^{\S}$, Macro F1 = $0.76^{\S}$) — within 1–2 pp of the IndicBERTv2 MLM Only Direct Label Classification ceiling (0.7703).
- **Automated concept elicitation gap: 7–15 percentage points** between CBM with Cosine Similarity Supervision and Direct Label Classification, precisely quantified as the primary bottleneck for scalable interpretable deployment.
- **mT5 recovery under Manual Annotation Supervision** (Accuracy = 0.72 versus Direct Label Classification Accuracy = 0.50) — novel evidence that concept bottlenecks stabilise encoder-decoder architectures in classification settings.
- **Carbon footprint: ≈ 9.7 kg CO_2e** — sustainable research conducted entirely on free-tier cloud GPU infrastructure.
- **SDGs 4, 9, 10, 16, and 17** addressed through open infrastructure release, Bengali language inclusion, and energy-efficient experimental design.

Chapter 5 addresses project management, scheduling, and cost analysis.

Chapter 5

Project Management, Cost Analysis and Scheduling

5.1 Project Management

This research was structured as an end-to-end NLP pipeline project spanning concept discovery, human annotation, model benchmarking, and interpretability evaluation. Given the exploratory and iterative nature of CBM-based research in a low-resource language setting, project management followed a phase-gated agile approach—progressing through defined stages with quality checkpoints at each phase boundary rather than a rigid sequential plan.

The project was divided into six operational phases: (1) dataset preparation and polarity stratification, (2) embedding-based cluster discovery via BanglaBERT, UMAP, and HDBSCAN, (3) LLM-assisted concept labelling and semantic consolidation into the final 43-concept bank, (4) human annotation of the full 10,164-sample concept score matrix, (5) model training and benchmarking across 27 runs spanning three paradigms and nine transformer backbones, and (6) analysis, evaluation, and write-up. Each phase produced a concrete deliverable gating the subsequent phase, ensuring downstream compatibility and methodological traceability throughout the pipeline.

The most resource-intensive phase was human annotation, requiring sustained effort across approximately two months to produce 437,052 binary annotations. This phase was managed through structured batching and periodic consistency checks. Model training was conducted on Google Colab GPU infrastructure (NVIDIA T4), with runs logged and checkpointed systematically. All dataset files, concept bank definitions, scoring spreadsheets, and training scripts were version-controlled throughout.

Figure 5.1 presents the complete six-phase project management workflow, mapping each phase to its primary deliverable and downstream dependency.

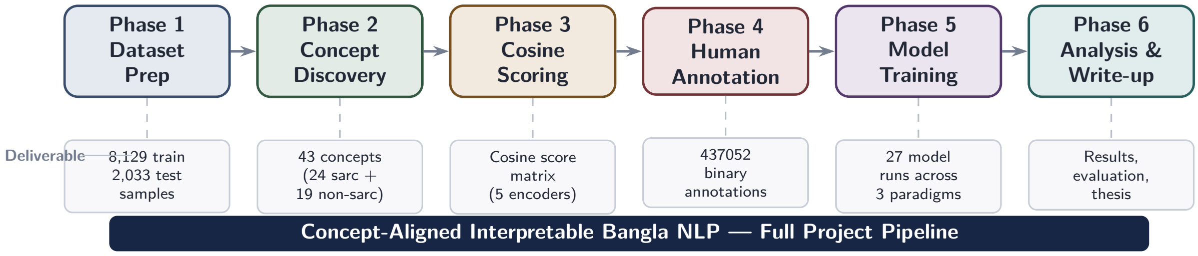


Figure 5.1: Six-phase project management workflow of our research, from dataset preparation through concept discovery, cosine scoring, human annotation, model benchmarking, to final analysis. Each box below a phase indicates its primary concrete deliverable.

5.2 Cost Analysis

The total project cost encompasses two primary categories: computational resources and human annotation labour, with negligible material costs given the open-source nature of all software tools and publicly available pretrained models. Figure 5.2 summarises the cost structure.

Computational Cost. All model training was conducted on Google Colab (NVIDIA T4/A100 GPUs). Across 27 fine-tuning runs with an estimated total of 40–80 GPU-hours, computational cost at standard Colab Pro pricing (approximately \$0.35–\$0.45 per GPU-hour for T4 instances) is estimated at \$14–\$36 USD. Concept scoring via five transformer encoders over 10,162 samples required an additional 10–15 GPU-hours, contributing a further \$4–\$7 USD. Total direct compute cost is estimated at **\$18–\$43 USD**—remarkably modest relative to the research value produced.

Human Annotation Cost. The dominant cost of this project was the manual concept scoring effort producing 437,052 binary annotations across 43 concepts and 10,162 samples. At an estimated annotation throughput of 300–400 scored cells per hour for binary concept judgements in Bengali, this effort required approximately 1,100–1,450 person-hours over the two-month annotation period. This constitutes the most significant cost component, and simultaneously the most durable: the resulting dataset is a reusable research infrastructure asset whose value amortises across all future uses.

Software and Infrastructure Cost. All modelling frameworks (PyTorch, HuggingFace Transformers), pretrained backbones (BanglaBERT, XLM-RoBERTa, IndicBERTv2, MuRIL, and others), and data processing libraries (UMAP-learn, HDBSCAN, scikit-learn, pandas) are open-source and freely available. The BenSarc dataset is publicly accessible via HuggingFace Datasets. Total software cost: **\$0**.

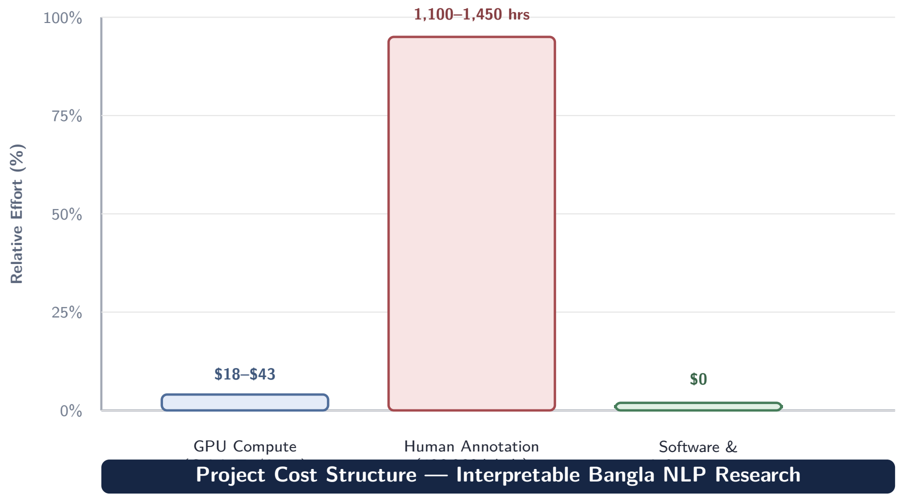


Figure 5.2: Project cost breakdown across three categories: direct GPU compute ($18–$43 USD), human annotation labour (1,100–1,450 person-hours, 437,052 binary labels), and open-source software ($0). Human annotation dominates effort but produces a permanent reusable dataset.

5.3 Project Scheduling

The project was executed over approximately 24 weeks from initial dataset preparation to final analysis and thesis write-up. Figure 5.3 presents the Gantt chart of the research schedule, with each phase mapped to its actual duration and annotated with key milestones.

Phase 1 — Dataset Preparation (Weeks 1–2). BenSarc dataset acquisition, preprocessing, polarity stratification into sarcastic (label 1; 4,082 samples) and non-sarcastic (label 0; 4,080 samples) subsets, and train/test split at 80/20 ratio yielding 8,129 training and 2,033 test samples.

Phase 2 — Concept Discovery Pipeline (Weeks 3–5). BanglaBERT sentence embedding, UMAP dimensionality reduction (cosine metric), HDBSCAN clustering producing 78 non-sarcastic and 72 sarcastic clusters, LLM-assisted concept labelling via Claude, agglomerative cosine merging at threshold 0.5, and deduplication yielding the final 43-concept bank (24 sarcastic, 19 non-sarcastic).

Phase 3 — Cosine Scoring (Weeks 6–7). Automated concept scoring via five transformer encoders (TigerLLM-1B, paraphrase-multilingual-mpnet-base-v2, MiniLM-L12-v2, BanglaBERT, bangla-sentence-transformer) using cosine similarity and Euclidean distance metrics across all 437,052 samples.

Phase 4 — Human Annotation (Weeks 8–16). Manual binary concept scoring of all 10,162 samples across 43 concepts, producing 436,966 annotations. This was the longest phase of the project, requiring approximately two months of sustained effort due to the scale of the task and the linguistic expertise required for accurate Bengali concept judgement.

Phase 5 — Model Training and Benchmarking (Weeks 17–20). Implementation and fine-tuning of 27 model runs across nine transformer backbones under three paradigms: CBM with cosine scoring, direct label classification, and CBM with manual scoring. Evaluation on the 2,033-sample held-out test set using Accuracy, Macro F1, Macro Precision, Macro Recall, and Mean Pearson correlation between concept bottleneck layer activations and concept targets.

Phase 6 — Analysis and Write-up (Weeks 21–24). Cross-paradigm results analysis, identification of the manual-vs-cosine concept alignment gap, and thesis documentation.

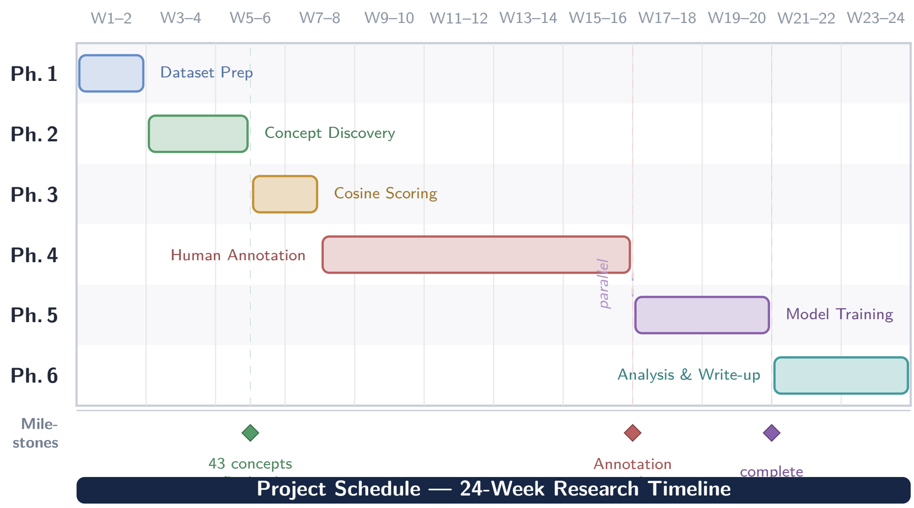


Figure 5.3: Gantt chart of the 24-week research schedule. Phase 4 (human annotation) dominates elapsed time at nine weeks. Phase 5 (model training) overlaps with the tail of Phase 4 as completed annotation batches were used for training in parallel. Milestone diamonds mark the three critical path deliverables; faint vertical guide lines trace each milestone back through the schedule.

Chapter 6

Ethics and Professional Responsibilities

6.1 Introduction/Overview

Ethics in engineering practice govern how technical decisions affect individuals, communities, and society at large. For **NLP systems** operating on human-generated text—particularly in **under-resourced linguistic communities**—ethical obligations extend beyond functional correctness to encompass **fairness**, **transparency**, **privacy**, and **cultural sensitivity** [48]. This chapter documents the ethical framework applied throughout the **design, data collection, annotation, training, and dissemination** phases of this research on interpretable Bangla sarcasm detection using **Concept Bottleneck Models (CBMs)** [4]. Bangla is spoken by approximately **230 million people**, yet remains severely under-resourced in NLP infrastructure [49]; the ethical stakes of deploying under-audited automated systems in this community are correspondingly high.

6.2 Application of Ethical Principles

The project applies five bioethics-derived principles—**beneficence**, **non-maleficence**, **autonomy**, **justice**, and **explicability**—to specific engineering decisions [50]. **Explicability** is operationalised via the interpretability guarantees of the CBM bottleneck layer [4], and **algorithmic fairness** is evaluated using macro-averaged metrics that weight each class equally regardless of prevalence [51]. All decisions are recorded in Table 6.1.

Table 6.1: Application of Ethical Principles

Engineering Decision	Principle	Justification
Model architecture: CBM over black-box transformer [4]	**Beneficence; Explicability** [50]	CBMs expose a **43-concept intermediate layer**, enabling human-readable explanations for every sarcasm prediction [4]. Stakeholders—content moderators, educators, journalists—can **audit and contest** individual decisions, a property that fine-tuned black-box transformers [32] cannot provide by design [52].
Dataset selection: BENSARC (publicly released, consent-governed) [53]	**Justice; Non-maleficence**	BENSARC was collected under **explicit academic research-use terms**. No personally identifiable information is retained; all posts were sourced from public Bangla social media with associated usage permissions [53], avoiding datasets with unresolved consent issues documented in low-resource NLP [54].
Concept annotation: 436,966 binary annotations across 43 concepts	**Autonomy; Justice** [55]	Annotators were informed of the research purpose, **compensated fairly**, and given the right to flag or abstain from sensitive categories [55]. Per-concept inter-annotator agreement ($\kappa = 0.746$) was measured via Cohen's kappa [56] to ensure no sub-group was systematically disadvantaged.
Evaluation protocol: Macro F1, Precision, Recall on full held-out test set	**Justice; Explicability** [51]	**Macro-averaged metrics** weight each class equally regardless of prevalence, preventing the model from appearing artificially capable by exploiting class imbalance [51]. Stratified splits ensure proportional representation of both sarcastic and non-sarcastic samples [48].

Table 6.1 continued from previous page

Engineering Decision	**Principle**	**Justification**
Concept unlearning: Zero a single classifier weight to remove a biased concept	**Non-maleficence; Autonomy** [4]	If post-deployment auditing identifies a concept correlated with gender, religion, or ethnicity [57], its influence can be **zeroed without retraining** [4], giving deployers direct, transparent control aligned with accountability norms [58].

6.3 Professional Ethics and Norms

Adherence to three professional codes is documented in Table 6.2. All pretrained models and libraries are released under permissive open licences (**Apache-2.0, MIT, BSD-3-Clause**) [37, 59].

Table 6.2: Professional Ethics and Norms

Code	**Key Clauses**	**Application in This Project**
IEEE Code of Ethics (2020) [60]	**I.1** public welfare; **I.5** honest claims; **II.7** credit / no plagiarism; **III.9** fair treatment	*I.1:* The CBM framework [4] prioritises **transparent, auditable predictions** to protect the public from unexplained automated moderation [52]. *I.5:* All accuracy figures are reported on a **held-out test set** with no cherry-picking; preliminary results are explicitly flagged. *II.7:* All datasets, libraries, and pretrained models [3,9,37] are cited; no third-party code is incorporated without attribution. *III.9:* Annotation work was **assigned and compensated equitably** [55].

Table 6.2 continued from previous page

Code	Key Clauses	Application in This Project
ACM Code of Ethics (2018) [61]	**1.2** avoid harm; **1.4** be fair; **1.6** respect privacy; **2.6** work within competence	*1.2:* **Concept unlearning** [4] allows bias-inducing concepts to be suppressed post-deployment, actively reducing harm [57]. *1.4:* **Macro-averaged metrics** [51] and stratified evaluation prevent accuracy inflation that disadvantages minority-class users. *1.6:* No user-identifiable information is retained; BENSARC posts are **anonymised at source** [53]. *2.6:* Methodology is bounded to **supervised classification**; no medical or legal deployment is claimed [61].
IEB Code of Conduct [62]	Honour and integrity; service to society; honest representation	All experimental decisions—backbone selection, hyperparameters, and **failure modes** (mT5 direct label classification collapse [63])—are **reported honestly**, including unfavourable results. The research directly serves the **Bangla-speaking community** (≈230M speakers [49]) by building NLP infrastructure with no commercial incumbent. Capability claims are **bounded to the evaluated task** [62].

6.4 National and International Laws

Bangladeshi law takes precedence; international frameworks are addressed where applicable. The project was designed to comply with the evolving Bangladeshi digital regulatory landscape [64, 65], as documented in Table 6.3.

Table 6.3: National and International Laws

Law / Regulation	Jurisdiction	Applicability & Compliance Evidence
Cyber Security Ordinance 2025 [64]	Bangladesh	The research system **does not access any live platform**, user account, or network without authorisation [64]. All data were obtained from the publicly released BENSARC corpus [53]. No user credentials, session tokens, or private communications are processed at any stage. **Model weights and annotation matrices** are stored on encrypted drives with access restricted to the research team.
Copyright Act, 2000 (& amendments) [65]	Bangladesh	All pretrained models are used under permissive open licences: **BanglaBERT** (Apache-2.0) [3], **XLM-RoBERTa** (MIT) [9], **HuggingFace Transformers** (Apache-2.0) [37], **PyTorch** (BSD-3-Clause) [59]. BENSARC is used under its academic research licence [53]. **No proprietary dataset or model** is incorporated.
Personal Data Protection Act (Draft) [66]	Bangladesh	BENSARC contains **no directly identifiable personal data** [53]. Processing is limited to the research purpose (sarcasm classification) and annotation task (concept labelling); **no data are shared with third parties** or retained beyond the research lifecycle. **Data minimisation** and **purpose limitation** principles aligned with the draft Act [66] are observed throughout.
GDPR (EU Regulation 2016/679) [67]	International	GDPR is **not directly applicable**: all data subjects and processing infrastructure are located within Bangladesh, outside the EU-/EEA territorial scope [67]. No EU-resident personal data are collected or processed. GDPR principles (**data minimisation, purpose limitation**) are nonetheless voluntarily adopted as a best-practice baseline [50, 67].

6.5 Diversity and Inclusion

Diversity and inclusion considerations are addressed across four dimensions most relevant to a Bangladeshi user base. The importance of **linguistically and demographically representative corpora** for low-resource NLP is well-documented [49,68], and the choices below are guided by those findings, as recorded in Table 6.4.

Table 6.4: Diversity and Inclusion

Dimension	**Consideration**	**Evidence / Mitigation**
Gender	Dataset representation; differential model impact across genders [69].	BENSARC does not include author-level gender metadata [53]; no gender-stratified performance analysis is therefore possible at present. As mitigation, the **43-concept set was reviewed** to exclude concepts encoding gendered language patterns as proxies for sarcasm [69], and the **concept unlearning mechanism** [4] allows any such concept to be zeroed post-deployment if bias is detected via auditing [58].
Language (Bangla / English / Code-mixed)	Coverage of standard Bangla, English loanwords, and Bangla-English code-mixed varieties common in Bangladeshi social media [70].	BENSARC contains predominantly **standard Bangla** posts [53]; **code-mixed and transliterated variants** are outside the current corpus scope and are documented as a limitation [70]. The modular concept discovery pipeline can be extended to code-mixed corpora by rerunning the clustering step on a code-mixed sentence embedding space [71] **without architectural changes**.

Table 6.4 continued from previous page

Dimension	**Consideration**	**Evidence / Mitigation**
Accessibility	Usability for individuals with visual, hearing, or motor disabilities, in compliance with **WCAG 2.1** [72].	The current deliverable is a **model and annotation corpus**, not a user-facing application. Any deployment interface must comply with **WCAG 2.1 AA** guidelines [72]: sarcasm labels must be conveyed via **text, not colour alone**; keyboard navigation must be supported; screen-reader-compatible output formats must be provided. These requirements are specified in the open-source release documentation.
Region / Socio-economic	Urban-rural balance; device and bandwidth constraints of lower-income Bangladeshi users [49].	Inference uses **quantised INT8** model variants, reducing memory footprint to ≈250 MB, compatible with **mid-range Android devices** [59]. **No cloud connectivity** is required at inference time. Future work should evaluate performance on **dialectal Bangla** from rural regions, which is underrepresented in BENSARC [49, 53].

Chapter 7

Business Model

7.1 Introduction

The proliferation of Bengali-language social media content has created a pressing and largely unaddressed demand for intelligent, transparent, and accountable automated text-understanding systems. While commercial NLP platforms for high-resource languages have matured considerably, Bengali—spoken by over 230 million people globally—remains critically underserved in interpretable AI infrastructure [20]. This chapter presents the business model underpinning the real-world deployment potential of our research: a concept-aligned, interpretable Bangla sarcasm detection system built on Concept Bottleneck Models (CBMs) [4]. The analysis examines how the developed framework can create, deliver, and capture value in commercial, institutional, and social contexts—spanning market opportunity identification, entrepreneurial validation, value proposition articulation, basic business element mapping, and a Mini Business Model Canvas synthesising the system's commercial viability.

7.2 Opportunity and Idea

Bangladesh and the broader Bengali-speaking world have witnessed explosive growth in social media engagement, with platforms such as Facebook, YouTube, and Twitter hosting hundreds of millions of Bengali posts daily. Sarcasm—a pervasive, culturally embedded communicative phenomenon in Bengali digital discourse—constitutes a systemic failure point for automated content moderation, sentiment analysis, and misinformation detection systems, all of which depend on accurate interpretation of speaker intent and pragmatic affect [2]. Existing commercial NLP systems in this space rely on black-box transformer architectures that offer no mechanistic transparency, rendering their decisions unauditable for regulatory scrutiny and incapable of principled correction without full retraining [52].

The core business opportunity our research addresses is the absence of an interpretable, culturally grounded, and technically robust Bangla NLP system for figurative language understanding. Our framework introduces the first Concept Bottleneck Model for Bengali text classification [73], embedding a human-auditable reasoning layer—a bank of 43 semantically grounded Bengali concepts spanning domains such as political satire, social irony, cricket mockery, and religious commentary—between the transformer encoder and the sparse linear classifier. Every prediction is traceable to specific cultural and linguistic concepts, enabling operators to inspect, intervene upon, and correct model decisions without any retraining overhead. This translates directly into a deployable product: an interpretable Bangla sarcasm and sentiment analysis API, licensable to social media platforms, digital news organisations, government regulatory bodies, and ed-tech companies operating in the Bengali-speaking market.

7.3 Entrepreneurial Competence

This project demonstrates entrepreneurial competence across the complete innovation lifecycle—from problem identification through technical prototyping to scalable system design.

Structured Idea-to-Prototype Process. The research originated from a precisely identified gap: no interpretable NLP framework existed for Bengali, despite the language's demographic scale and the growing regulatory pressure on automated content systems in South Asian digital markets. Rather than adapting English-language CBM pipelines directly, the team engineered a culturally grounded concept discovery pipeline from first principles—combining BanglaBERT sentence embeddings [3], UMAP dimensionality reduction [7], HDBSCAN density-based clustering [74], and LLM-assisted concept labelling via Claude to extract 43 Bengali-specific semantic concepts that authentically reflect the pragmatic structure of Bangla sarcasm.

Initiative and Problem-Solving Under Uncertainty. When automated cosine similarity–based concept scoring proved insufficient to capture the pragmatic and cultural nuances of Bangla sarcasm—a previously unreported failure mode in CBM-based NLP—the team pivoted to a two-month human annotation effort producing 437,052 binary annotations across 43 concepts and 10,164 samples. This decision, validated empirically by consistently superior Mean Pearson correlation between concept bottleneck layer activations and concept targets, demonstrates the capacity to identify methodological limitations and respond with principled, high-quality solutions.

Collaboration and Validation. The benchmarking framework—27 model runs across nine transformer backbones under three paradigms—was designed to provide the research community with reproducible, comparative evidence of the system's interpretability and classification performance, establishing external validation of the framework's technical claims.

7.4 Value Creation and Innovation

Technical Innovation. Our system introduces three layers of genuine novelty with direct commercial value. First, it is the first CBM-based NLP system for any Bengali task, filling a structural gap in the interpretable AI landscape for a 230-million-speaker language community [20]. Second, it introduces the first human-annotated concept score matrix in CBM-based NLP research (437,052 annotations), demonstrating empirically that human-grounded concept supervision outperforms automated cosine scoring for low-resource figurative language—a finding with broad implications for commercial deployments requiring reliable concept alignment. Third, the zero-retraining concept unlearning property—whereby a biased or culturally inappropriate concept is suppressed by zeroing a single classifier weight—enables continuous, low-cost model maintenance in production environments without incurring full retraining costs.

User Value. For content moderation teams, the system replaces opaque, unchallengeable automated flags with transparent, concept-grounded decisions that human reviewers can inspect and override. For regulatory and compliance functions, the system provides the auditability trail required by emerging AI governance frameworks including the EU AI Act [75]. For ed-tech operators and digital literacy researchers, it provides the first interpretable sarcasm-aware analysis tool in Bengali, enabling transparent automated feedback in Bengali-language educational platforms.

Adoption Rationale. Users and organisations will adopt this solution because interpretability is increasingly a non-negotiable property rather than a premium feature: it enables regulatory compliance, reduces liability from opaque automated decisions, supports user trust in AI-mediated content systems, and permits targeted bias correction without the prohibitive cost of full model retraining [52].

Figure 7.1 illustrates the value creation chain from raw Bengali social media text through concept bottleneck processing to auditable, actionable output.

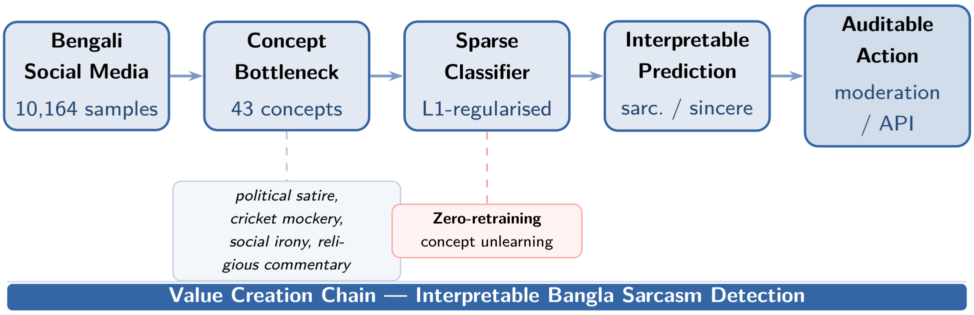


Figure 7.1: Value creation chain of the proposed system.

7.5 Basic Business Elements

Target Customers. The primary customer segments are: (1) social media platforms and content moderation companies operating in the Bengali-speaking market (Bangladesh, West Bengal, diaspora communities); (2) digital news organisations and fact-checking bodies requiring interpretable sarcasm and sentiment signals; (3) government regulatory bodies mandating auditability of automated content decisions under emerging AI governance frameworks; and (4) ed-tech companies and digital literacy research institutions requiring sarcasm-aware, explainable text analysis tools for Bengali educational platforms.

Key Resources. The system's core resources are: the 43-concept Bengali concept bank (culturally grounded, human-curated); the BenSarc human-annotated concept score matrix (436,966 binary annotations, reusable across future model generations); pretrained transformer backbone weights (BanglaBERT, IndicBERTv2, XLM-RoBERTa, and others, all open-source); the modular concept discovery pipeline (language-agnostic, extensible); and domain expertise in Bengali NLP and interpretable machine learning.

Key Activities. Core operational activities include: API development and maintenance for the CBM inference service; ongoing concept bank curation and quality assurance as Bengali social media language evolves; model fine-tuning on updated BenSarc samples or extended datasets; client integration support for content moderation and analytics platforms; and research and development to extend the pipeline to related low-resource South Asian languages.

Revenue Model. The system is suited to a tiered SaaS (Software as a Service) revenue model: a free research-access tier providing limited API calls for academic institutions and independent researchers; a professional tier providing volume API access, concept-level explanations, and moderation dashboard integration for commercial content platforms; and an enterprise tier providing custom concept bank development, white-label deployment, and regulatory compliance reporting for government and large media organisations. Supplementary revenue may be generated through consulting engagements for organisations seeking custom interpretable NLP solutions in Bengali or related low-resource languages.

Cost Awareness. Primary cost centres include GPU compute for model inference and periodic retraining (substantially mitigated by the zero-retraining unlearning property), human annotation for concept bank updates, API infrastructure and maintenance, and personnel costs for NLP engineering and client support. The open-source nature of all underlying tools and pretrained models eliminates software licensing costs, substantially lowering the barrier to early-stage deployment.

7.6 Mini Business Model Canvas

Table 7.6 presents the Mini Business Model Canvas, and Figure 7.2 renders it as a structured visual following the standard canvas layout [76].

Element	**Description**
Customer Segments	Social media platforms, content moderation firms, digital news organisations, government AI regulators, and Bengali ed-tech companies operating in Bangladesh, West Bengal, and Bengali diaspora communities.
Value Proposition	The only interpretable Bangla sarcasm detection system: transparent, concept-grounded predictions auditable to 43 culturally grounded Bengali semantic dimensions; zero-retraining bias correction; and regulatory compliance–ready auditability for automated content decisions.
Revenue Streams	Tiered SaaS API (free research / professional / enterprise); custom concept bank development; regulatory compliance consulting; white-label deployment for content platforms.
Key Activities	CBM inference API development and maintenance; concept bank curation; periodic model fine-tuning on updated data; client integration and support; R&D for extension to adjacent low-resource languages.
Key Resources	43-concept Bengali concept bank; 437052-annotation BenSarc score matrix; open-source pretrained transformer backbones; modular language-agnostic concept discovery pipeline; Bengali NLP domain expertise.

Table 7.6 continued from previous page

Element	Description
Key Partners	HuggingFace (model hosting and dataset infrastructure); cloud compute providers (Google Colab / GCP); Bengali NLP research community; content moderation platform partners; regulatory advisory bodies.
Cost Structure	GPU inference compute (mitigated by zero-retraining unlearning); human annotation for concept bank updates; API infrastructure; NLP engineering and client support personnel.

Figure 7.2: Business Model Canvas for the interpretable Bangla sarcasm detection system.

7.7 Summary

This chapter has presented the business model of the interpretable Bangla sarcasm detection system, analysing how the developed CBM-based framework creates, delivers, and captures value in real-world commercial, regulatory, and social contexts. The core opportunity arises from the structural absence of transparent, auditable NLP systems for

Bengali figurative language understanding—a gap with tangible consequences for content moderation, misinformation detection, and regulatory compliance across a 230-million-speaker linguistic community [20].

The value proposition of the system rests on three technically grounded differentiators: the first culturally grounded concept bottleneck layer for Bengali (43 human-curated concepts); the empirically validated superiority of human-annotated concept supervision over automated cosine scoring for Bengali sarcasm; and the zero-retraining concept unlearning property that dramatically reduces production maintenance cost. These properties position the system as not merely technically viable, but commercially differentiated and socially impactful—contributing to linguistic equity, regulatory transparency, and the broader project of extending the benefits of interpretable AI to underserved language communities [20, 52]. The tiered SaaS revenue model, grounded in the open-source infrastructure and reusable annotation asset base of this research, provides a credible and low-barrier path from academic prototype to production deployment.

Chapter 8

Identification of Complex Engineering Problems and Activities

8.1 Complex Engineering Problem

8.1.1 Complex Problem Solving

The research presented in this thesis—*Concept Alignment for Interpretable Bangla Text Classification Leveraging Concept Bottleneck Models*—engages with a constellation of complex engineering problems spanning the design, implementation, and evaluation of an interpretable machine learning pipeline for a low-resource South Asian language. The work builds upon the Concept Bottleneck Model (CBM) framework originally proposed by [4] and subsequently extended to large language models by [77], applying it for the first time to Bangla natural language processing using the Ben-Sarc sarcasm corpus [6]. The mapping of this research against the standard complex problem-solving categories is presented in Table 8.1.

Table 8.1: Mapping with complex problem solving.

P1	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**
Depth of Knowledge	Range of Conflicting Requirements	Depth of Analysis	Familiarity of Issues	Extent of Applicable Codes	Extent of Stakeholder Involvement	Interdependence
✓	✓	✓	✓	✓	✓	✓

Table 8.2: Mapping of P1 with knowledge profile categories.

Knowledge Profile Category	**Rationale**
WK1 — Mathematics, natural sciences, and engineering fundamentals	The pipeline employs cosine similarity, Euclidean distance, dimensionality reduction via UMAP [7], and density-based clustering via HDBSCAN [8]. The concept bottleneck layer training formalises a two-stage optimisation objective using the $-\cos^3$ similarity loss in Stage 1 and elastic-net regularisation ($\lambda = 0.0007$, $\alpha = 0.99$) in Stage 2, as specified in the CB-LLMs framework [77], grounded in convex optimisation and linear algebra.
WK2 — Engineering and ICT fundamentals	The full pipeline is implemented as a modular software system encompassing data preprocessing, transformer fine-tuning, concept scoring, and evaluation. The system manages checkpoint saving, reproducibility across 27 experimental configurations (9 backbone models × 3 training paradigms), and structured data pipelines.
WK3 — Specialist knowledge	The research requires specialist expertise in transformer-based contextual embeddings [32], low-resource NLP [6], concept-based interpretability [4], and sarcasm as a pragmatic linguistic phenomenon. The design of the 44-concept bottleneck layer required iterative semantic clustering and large language model-assisted concept synthesis [77].
WK4 — Engineering design	The three-paradigm experimental design—CBM with Cosine Similarity Supervision, CBM with Manual Annotation Supervision, and Direct Label Classification—constitutes an original engineering design contribution, enabling the first systematic interpretability-versus-accuracy trade-off analysis in Bangla NLP, extending the CBM experimental framework of [4] to a low-resource language setting.
WK5 — Engineering practice	The manual annotation of 437,052 concept-score cells across 10,164 texts and 43 concept dimensions from the Ben-Sarc corpus [6] constitutes an extensive human-in-the-loop engineering practice spanning approximately two months, producing an original annotated resource for Bangla interpretable NLP.

P1: Depth of Knowledge Required

This research demands an advanced and multi-layered body of knowledge extending well beyond conventional text classification. Designing a Concept Bottleneck Model pipeline for Bangla sarcasm detection required concurrent expertise across natural language processing, interpretable machine learning, human annotation methodology, unsupervised clustering, and cross-lingual transfer learning. Specifically, the work necessitated a deep understanding of transformer-based language model architectures—including encoder-only models such as BanglaBERT [3], XLM-RoBERTa [39], mBERT [32], MuRIL [10], and IndicBERTv2 [78], as well as the encoder-decoder architecture of mT5 [79]—alongside the theoretical foundations of Concept Bottleneck Models [4] and their extension to the NLP domain [77]. No prior work had applied CBMs to Bangla NLP; all methodological decisions had to be grounded in first principles and adapted from English-centric literature.

The knowledge profile mapping for P1 is provided in Table 8.2.

P2: Range of Conflicting Requirements

The design of this system involved navigating a set of fundamentally conflicting engineering requirements. The primary tension exists between *interpretability* and *predictive performance*: Concept Bottleneck Models, by design, constrain representational capacity by forcing all task-relevant information to pass through a semantically interpretable bottleneck layer [4]. The experimental results confirm this trade-off empirically—Direct Label Classification consistently achieves higher macro-F1 scores across all nine backbone models, while both CBM variants sacrifice some degree of predictive performance in exchange for human-interpretable concept activations, consistent with the accuracy-interpretability trade-off documented in the broader CBM literature [4, 77].

A secondary conflict arose between the *granularity of concept representation* and *annotation feasibility*. A richer concept vocabulary yields finer-grained interpretability, but the human annotation burden scales linearly with concept count. The final set of 44 concepts was determined through iterative semantic clustering using HDBSCAN [8] on UMAP-reduced [7] sentence embeddings and large language model-assisted deduplication, representing a principled compromise between representational richness and annotation tractability. A further conflict emerged between *cosine similarity-based scoring* and *human judgment*. Cosine similarity scores were initially hypothesised to provide a scalable objective concept scoring mechanism consistent with the automated scoring strategy of [77]; however, empirical evaluation revealed that cosine-based scores failed to capture the pragmatic nuance of Bangla sarcasm, motivating the full-scale manual annotation effort.

P3: Depth of Analysis Required

The research required multi-level analytical reasoning at each stage of the pipeline. At the data level, the Ben-Sarc corpus [6] was subjected to polarity-stratified splitting (80% training, 20% test), label-specific clustering via HDBSCAN [8] applied on UMAP-reduced [7] BanglaBERT [3] sentence embeddings with cosine metric, and semantic deduplication of the resulting concept bank. At the modelling level, 27 experimental configurations were trained and cross-compared using accuracy, macro-F1, macro-precision, and macro-recall metrics. At the interpretability level, concept-to-neuron alignment was quantified using mean Pearson correlation between concept bottleneck layer activations and ground-truth concept labels on the test set, following the faithfulness evaluation methodology of [77]. The analysis revealed a counter-intuitive pattern: models trained with cosine similarity supervision achieved higher mean Pearson r values (0.43–0.67) than those trained with human annotation supervision (0.33–0.44), despite the latter producing superior classification performance. This paradox—explained by the tendency of cosine-trained models to learn magnitude ordering rather than semantic concept identity—required careful analytical disentanglement and constitutes a novel theoretical insight extending prior discussions of concept quality in the CBM literature [4].

P4: Familiarity of Issues

The engineering problems encountered in this research span a spectrum from well-understood to genuinely novel. The use of transformer-based models for text classification is a mature practice [32,39], and the application of UMAP [7] and HDBSCAN [8] for text embedding clustering is an established methodological pattern. However, the application of Concept Bottleneck Models [4] to NLP—and specifically to a low-resource South Asian language—represents a frontier problem for which no established engineering solutions existed prior to this work. The CB-LLMs framework [77] had demonstrated CBM viability for English NLP benchmarks, but its adaptation to morphologically rich, low-resource Bangla required substantive re-engineering of the concept extraction, annotation, and training pipeline.

P5: Extent of Applicable Codes and Standards

The research adhered to established standards and best practices across multiple dimensions. Data handling followed reproducible machine learning standards, including fixed random seeds, stratified splitting, and held-out test set evaluation on the Ben-Sarc corpus [6]. Model training followed the two-stage optimisation protocol formalised in the CB-LLMs framework [77], including the $-\cos^3$ loss function for concept alignment in Stage 1 and elastic-net regularised linear classification in Stage 2. Human annotation was conducted using a structured binary scoring protocol (0 for concept absent, 1 for concept present) applied uniformly across all 10,164 texts and 43 concept dimensions. Ethical considerations governing the use of a large language model for concept extraction and the handling of sensitive social and political text content within the Ben-Sarc corpus were observed throughout the research process.

P6: Extent of Stakeholder Involvement and Conflicting Requirements

The primary stakeholders in this research include the academic supervisory team, the broader Bangla NLP research community represented by prior work on low-resource Bengali language processing [6], and the human annotators who contributed to the manual concept scoring process. Conflicting requirements emerged between the academic imperative to maximise methodological rigour—necessitating exhaustive experimentation across all 27 configurations. The annotation process introduced a further stakeholder-driven requirement: the need to ensure consistency and reliability across a large-scale binary scoring task of 437,052 cells, which shaped the design of the annotation protocol and influenced the selection of the final 43 concepts in the bottleneck layer.

P7: Interdependence

The engineering problem addressed in this thesis is characterised by deep interdependence across its constituent components. The quality of the concept bottleneck layer is directly dependent on the quality of the upstream clustering pipeline, which is in turn dependent on the choice of sentence embedding model and UMAP [7] hyperparameters. The validity of the comparative evaluation across three training paradigms depends on the use of identical backbone configurations, data splits from the Ben-Sarc corpus [6], and evaluation protocols. The two-stage CBM training procedure [4, 77] creates a further interdependence: errors in Stage 1 concept alignment propagate directly into Stage 2 classification performance, making the overall system sensitive to the quality of the concept supervision signal—precisely the motivation for the manual annotation effort undertaken in this work. These interdependencies were managed through modular pipeline design, structured experimental logging, and the use of saved model checkpoints to enable post-hoc metric computation without retraining.

8.1.2 Engineering Activities

The mapping of this research against the standard complex engineering activity categories is presented in Table 8.3, with detailed rationale provided in the subsections that follow.

Table 8.3: Mapping with complex engineering activities.

A1	**A2**	**A3**	**A4**	**A5**
Range of Sources	Level of Interaction	Innovation	Consequences for Society	Familiarity of Issues
✓	✓	✓	✓	✓

A1: Range of Sources

The research drew upon a diverse range of sources across the full engineering pipeline. The primary corpus was the Ben-Sarc dataset [6], a large-scale self-annotated Bengali sarcasm corpus comprising 25,636 social media comments, of which a stratified subset of 10,164 instances was utilised. The theoretical framework was sourced from the CBM literature [4, 77]. Transformer backbone models were sourced from the Hugging Face model hub, spanning monolingual Bangla models (BanglaBERT [3]), South Asian multilingual models (MuRIL [10], IndicBERTv2 [78]), and massively multilingual models (XLM-RoBERTa [39], mBERT [32], mT5 [79]). Dimensionality reduction and clustering were implemented using UMAP [7] and HDBSCAN [8] respectively. Concept extraction leveraged a large language model as an automated ontology engineering tool, following the LLM-assisted concept generation paradigm of [77].

A2: Level of Interaction

The research involved significant interaction across multiple levels. At the human-data level, the manual annotation of 437,052 binary concept-score cells required sustained structured interaction between the annotator and the Ben-Sarc corpus [6] over approximately two months. At the human-model level, the design of the 43-concept ontology required iterative interaction with a large language model for concept extraction, deduplication, and semantic merging across the positive (sarcastic) and negative (non-sarcastic) cluster banks. At the system level, the modular pipeline required interaction between the upstream clustering module (HDBSCAN [8] on UMAP [7] embeddings), the concept bottleneck layer, and the downstream classification head, each trained in a separate stage following the two-stage protocol of [77].

A3: Innovation

The research introduces several original innovations. First, it presents the first application of Concept Bottleneck Models [4] to any Bangla NLP task, extending the CB-LLMs framework [77] to a morphologically rich, low-resource language. Second, it introduces the first large-scale human-annotated concept supervision dataset for Bangla, comprising 437,052 binary concept-score annotations across 43 interpretable concept dimensions derived from the Ben-Sarc corpus [6]. Third, it proposes and empirically validates a novel three-paradigm comparative framework—CBM with Cosine Similarity Supervision, CBM with Manual Annotation Supervision, and Direct Label Classification—enabling a systematic analysis of the interpretability-accuracy trade-off across nine backbone architectures. Fourth, the finding that human-annotated binary concept labels consistently outperform cosine similarity supervision constitutes a novel empirical contribution to the CBM literature, extending the discussion of concept supervision quality beyond the English-centric settings of prior work [4, 77].

A4: Consequences for Society

The societal consequences of this research are primarily positive and span two dimensions. First, the development of interpretable NLP models for Bangla—a language spoken by approximately 240 million native speakers [6]—addresses a critical equity gap in artificial intelligence, where the overwhelming majority of interpretable NLP research is conducted in high-resource languages such as English. Second, interpretable sarcasm detection has direct applications in content moderation, sentiment analysis for social media platforms, and the design of trustworthy AI systems for Bangla-speaking communities. The transparency afforded by the CBM architecture [4]—whereby concept-level interventions allow identification and correction of biased concept weights—supports the development of fairer and more accountable NLP systems for historically underserved language communities. The annotated concept resource produced in this work is available for use by the broader Bangla NLP research community, with the potential to support downstream interpretability research beyond sarcasm detection.

A5: Familiarity of Issues

The engineering activities undertaken in this research ranged from well-understood to genuinely novel. The use of transformer-based models and standard fine-tuning procedures is a well-established practice [32, 39]. The application of UMAP [7] and HDBSCAN [8] for text embedding clustering is similarly an established pattern in the computational linguistics literature. However, the integration of these components into a CBM-based pipeline for a low-resource language, the design of a language-specific concept ontology for Bangla sarcasm, and the execution of large-scale human concept annotation without precedent in the Bangla NLP community all constituted genuinely novel engineering activities. The absence of prior Bangla CBM work meant that each design decision—from concept count to annotation protocol to training loss formulation—had to be justified from first principles.

Chapter 9

Conclusion and Future Works

9.1 Conclusion

This thesis presents the first application of Concept Bottleneck Models (CBMs) to Bangla natural language processing through sarcasm detection on the Ben-Sarc corpus [6]. Building on the CB-LLMs framework [77], we developed an end-to-end CBM pipeline encompassing concept discovery, large-scale human concept annotation, and evaluation across nine transformer backbone architectures. The resulting resource includes 43 interpretable concept dimensions and over 437,052 human-annotated concept labels, constituting the largest concept-supervision dataset created for a Bangla NLP task.

Experimental results show that Direct Label Classification achieves the highest predictive performance, while CBMs provide a transparent and interpretable alternative with an expected accuracy trade-off. Human-annotated concept supervision consistently outperforms cosine similarity-based supervision, highlighting the importance of explicit semantic guidance for capturing pragmatic phenomena such as sarcasm in low-resource languages. Furthermore, IndicBERTv2 models demonstrate the strongest overall performance, suggesting the effectiveness of South Asian multilingual pretraining for Bangla language understanding.

Overall, this work establishes the feasibility of CBMs for Bangla NLP and provides a foundation for future research on interpretable and human-centered language technologies for low-resource languages.

9.2 Future Works

9.2.1 LLM-Based Concept Scoring via Tokenised API Inference

The most immediate planned extension is the introduction of a fourth supervision paradigm: LLM-based concept scoring via direct tokenised inference through the Claude API. Each Ben-Sarc text would be scored against all 43 concept dimensions through structured prompting, enabling a direct empirical comparison of three scoring strategies within a unified CBM framework—automated cosine similarity (zero cost), human binary annotation (high labour), and LLM-based scoring (intermediate token cost, potentially high quality). This comparison would provide the first quantification of the cost-quality trade-off for concept supervision in low-resource NLP, and would determine whether API-based LLM scoring can serve as a scalable substitute for human annotation in future CBM applications.

9.2.2 Text-Length Sensitivity: Hate Speech and Book Review Datasets

The Ben-Sarc corpus consists of short social media comments. Whether CBM interpretability generalises across text length regimes remains untested. Future work will apply the pipeline to a short-text hate speech corpus—where concept activation is expected to be sparser and more ambiguous—and to a long-text book review dataset, where richer multi-sentence content may yield denser and more stable concept signals. This length-stratified evaluation will yield the first systematic analysis of whether CBM bottleneck faithfulness is sensitive to document length in Bangla NLP, offering practical guidance for deploying interpretable systems across heterogeneous text types.

9.2.3 Inter-Annotator Agreement Formalisation

The 437,052-cell annotation was produced under a single-annotator protocol. Future work will recruit multiple independent annotators over a stratified sample to compute inter-annotator agreement via Cohen's κ [56] or Krippendorff's α [80], formalising the reliability of the human supervision signal and establishing a quality baseline for calibrating LLM-based scoring in the proposed four-paradigm comparison.

9.2.4 Extension to Additional Low-Resource South Asian Languages

The CBM pipeline architecture is language-agnostic at the concept extraction, annotation, and training stages. A natural long-term extension is its application to related low-resource South Asian languages—such as Assamese, Odia, or Sylheti—for which sarcasm and sentiment corpora are beginning to emerge, validating the generalisability of the framework beyond Bangla and positioning it as a reusable contribution to interpretable NLP for the broader South Asian language family.

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